

Appendix: Proofs of Asymptotic Properties for FACE-HD

Sinian Zhang

A Asymptotic Normality of the Cross-Fitted Estimator

A.1 Setup and Notation

We analyze the asymptotic properties of the K_f -fold cross-fitted estimator for a single target-source pair (t, s) , derived using the two-level cross-fitting strategy described in the main manuscript's appendix (Section B). Let K_f be a fixed integer (e.g., 5 or 10). The data $\mathcal{D} = \mathcal{D}_t \cup \mathcal{D}_s$ is partitioned into K_f folds, $\mathcal{D} = \bigcup_{k=1}^{K_f} \mathcal{D}_k$, where $\mathcal{D}_k = \mathcal{D}_{t,k} \cup \mathcal{D}_{s,k}$. Let $N_k = |\mathcal{D}_k| \approx N/K_f$. For notational simplicity, we assume the folds are balanced within each site (e.g., N_t and N_s are divisible by K_f) so that $|\mathcal{D}_{t,k}| = N_t/K_f$ and $|\mathcal{D}_{s,k}| = N_s/K_f$ for all k ; the minor rounding effects in practice are $o_p(N^{-1/2})$ and do not affect the first-order results. For any function $f(D)$ of a single observation $D = (R, \mathbf{X}, A, Y)$, we write $\tilde{E}_k[f(D)] := \frac{1}{N_k} \sum_{i \in \mathcal{D}_k} f(D_i)$ for the empirical average over the combined k -th fold. Let $\mathcal{D}^{(-k)} = \mathcal{D} \setminus \mathcal{D}_k$ denote the out-of-fold data used for training nuisance models evaluated on fold k . The crucial aspect of cross-fitting is that the data used for training, $\mathcal{D}^{(-k)}$, is independent of the data used for evaluation, $\mathcal{D}_k$.

Let $\vartheta = (\boldsymbol{\alpha}^1, \boldsymbol{\gamma}_1)$ be the vector of nuisance parameters. For this fixed pair (t, s) we write $\boldsymbol{\alpha}^1 \equiv \boldsymbol{\alpha}_{t,s}^1$ and $\boldsymbol{\gamma}_1 \equiv \boldsymbol{\gamma}_{s,1}$. Let $\vartheta^* = (\boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)$ denote the population targets of the calibrated estimation step (Equations (50) and (51) in the main manuscript), and let $\hat{\vartheta}^{(-k)} = (\hat{\boldsymbol{\alpha}}_{t,s}^{1,(-k)}, \hat{\boldsymbol{\gamma}}_{s,1}^{(-k)})$ be the nuisance estimators trained on data $\mathcal{D}^{(-k)}$ using the two-level cross-fitting procedure. Here the superscript $(-k)$ denotes the *final* two-level cross-fitted nuisance estimators used for evaluating fold k (including any inner-fold averaging), and hence $\hat{\vartheta}^{(-k)}$ is independent of $\mathcal{D}_k$.

To ensure numerical stability and theoretical regularity (bounded weights), we define a truncation function $\mathcal{T}(x) = \text{sign}(x) \min(|x|, M_\tau)$, where $M_\tau = M_{\tau,N}$ is allowed to depend on N (notation suppressed). (We avoid the symbol $\tau(\cdot)$ to prevent collision with the ATE τ in the main manuscript.) The truncated weighting function is defined as:

$$w_\tau(\mathbf{X}; \boldsymbol{\gamma}) := \frac{I(A=1)}{e^{\mathcal{T}(\phi(\mathbf{X})^T \boldsymbol{\gamma})}}.$$

Define also the (untruncated) weight

$$w(\mathbf{X}; \boldsymbol{\gamma}) := \frac{I(A=1)}{e^{\phi(\mathbf{X})^T \boldsymbol{\gamma}}}.$$

Remark on Truncation / Clipping: The truncation function $\mathcal{T}$ is applied *only in the calibrated loss functions* (Equations (50) and (51) in the main manuscript) to ensure numerical stability during nuisance estimation; the estimators and influence functions below use the untruncated weight $w(\mathbf{X}; \boldsymbol{\gamma})$, consistent with the main manuscript. Under Assumption A.2(c), truncation is asymptotically inactive at the population targets (so $w_\tau(\mathbf{X}; \boldsymbol{\gamma}_1^*) = w(\mathbf{X}; \boldsymbol{\gamma}_1^*)$ and $\psi'(\mathcal{T}(\phi(\mathbf{X})^T \boldsymbol{\alpha}^{1,*})) = \psi'(\phi(\mathbf{X})^T \boldsymbol{\alpha}^{1,*})$ w.p.a.1), and hence the population first-order conditions induced by the calibrated losses are unchanged up to an $o(1)$ term.

In the software implementation, this corresponds to using a finite M_τ during nuisance optimization, while the default inference computations use $M_{\tau, \text{inference}} = \infty$ (i.e., no truncation inside the final DR estimating equation). In addition, low-level C++ kernels retain very broad machine-safety bounds (e.g., `WEIGHT_MIN/MAX`, `RATIO_CLIP`) to prevent overflow/underflow; these are numerical safeguards and are treated as asymptotically inactive under the stated moment/overlap conditions. Setting a finite $M_{\tau, \text{inference}}$ corresponds to the clipped estimating equation discussed below.

If, instead, one *also* clips the weights in the final AIPW/DR estimating equation (i.e., replaces w by w_τ inside $\hat{\mu}_{t,s}^1$), then the resulting estimator has a different influence function in finite samples. Under Assumption A.2(c) (clipping probability $o(N^{-1})$ at the population targets), the difference between the clipped and unclipped estimators is $o_p(N^{-1/2})$ and the first-order limit theory is unchanged; otherwise, the clipping-induced bias must be accounted for (see, e.g., Tan [2020]).

A.2 Implementation Conventions Used in the Proof Mapping

For clarity, we record the conventions used to align implementation details with the asymptotic statements:

- **Fold-wise centering of IF components.** Variance and covariance components are computed from centered fold-level influence-function vectors, consistent with the sample-moment definitions in the main manuscript.
- **Weight-estimation variance vs final inference variance.** The penalized quadratic variance objective is used for estimating aggregation weights, whereas final cross-fitted uncertainty is computed from the sample-level unified estimating equation.
- **Finite-sample convexity stabilization.** In finite samples, the plug-in covariance matrix in the aggregation objective can be nearly indefinite; implementation-level symmetrization and minimal ridge stabilization are treated as numerical devices that vanish asymptotically under the regularity conditions.

Assumption-to-Implementation Map. Table 1 provides a concise correspondence between core assumptions/claims and implementation controls.

Theoretical condition	Role in proof	Implementation anchor
Fold-wise centering of IF components	Consistent variance/covariance moments	Fold-level centering in <code>R/cross_fitting_ag</code>
Training-only truncation (M_τ)	Nuisance stability, same first-order target	Calibrated losses in <code>R/cross_fitting_algor</code>
Inference untruncated by default	Matches unclipped IF expansion	<code>M.TAU_INFERENCE_DEFAULT = Inf</code> in <code>R/cons</code>
Finite-sample QP convexity handling	Well-posed optimization in practice	Symmetrize + PSD ridge in <code>src/weight_op</code>

Table 1: Assumption-to-implementation correspondence for reproducible inference.

We assume $\phi(\mathbf{X})$ includes an intercept term. Let $p := \dim(\phi(\mathbf{X}))$. For canonical exponential-family GLMs, we use the standard notation that the negative log-likelihood takes the form $-Y\theta + \Psi(\theta)$ with linear predictor $\theta = \phi(\mathbf{X})^T \alpha$, and that $\psi(\theta) = \Psi'(\theta)$; $\psi'(\theta)$ denotes the derivative with respect to θ . We also use the shorthand $\psi(\phi(\mathbf{X}); \alpha) \equiv \psi(\phi(\mathbf{X})^T \alpha)$. To avoid confusion with the notation used in the reference PDFs (in the `proof_ref` folder), we consistently follow the main manuscript’s notation: $\phi(\mathbf{X})$ denotes the feature/basis expansion and $\psi(\cdot)$ denotes the GLM conditional mean. (For example, some references use $\phi(\cdot)$ and $\Phi(\cdot)$ for a Normal density/CDF; we do *not* use that convention here.)

As in the main manuscript, the target estimand is the target-site treated potential outcome mean $\mu^1 \equiv \mathbb{E}_t[Y(1)]$. The K_f -fold cross-fitted estimator for μ^1 is:

$$\hat{\mu}_{t,s}^{1,\text{CF}} = \frac{1}{K_f} \sum_{k=1}^{K_f} \left\{ \tilde{E}_{t,k} [\psi(\phi(\mathbf{X}); \hat{\alpha}_{t,s}^{1,(-k)})] + \tilde{E}_{s,k} [w(\mathbf{X}; \hat{\gamma}_{s,1}^{(-k)}) (Y - \psi(\phi(\mathbf{X}); \hat{\alpha}_{t,s}^{1,(-k)}))] \right\}.$$

Here,

$$\tilde{E}_{t,k}[g(D)] := \frac{1}{N_{t,k}} \sum_{i \in \mathcal{D}_{t,k}} g(D_i), \quad \tilde{E}_{s,k}[g(D)] := \frac{1}{N_{s,k}} \sum_{i \in \mathcal{D}_{s,k}} g(D_i),$$

denote sample averages over the k -th fold of the target and source data, with $N_{t,k} = |\mathcal{D}_{t,k}|$, $N_{s,k} = |\mathcal{D}_{s,k}|$, and $N_k = N_{t,k} + N_{s,k}$. Let $N = N_t + N_s$ be the total sample size, and let $\rho_t = N_t/N$ and $\rho_s = N_s/N$. We write $\mathbb{E}[\cdot]$ for expectation under the pooled distribution, i.e., $\mathbb{E}[g(D)] = \rho_t \mathbb{E}_t[g(D)] + \rho_s \mathbb{E}_s[g(D)]$. Throughout the asymptotics we consider the standard two-sample (multi-sample) regime with sampling fractions bounded away from 0 and 1, i.e., $N_t/N \rightarrow \rho_t \in (0, 1)$ and $N_s/N \rightarrow \rho_s \in (0, 1)$. Under overlap/positivity for the relevant treatment arm, the effective sample sizes within each arm also scale linearly with N (e.g., $\sum_{i \in \mathcal{D}_s} I(A_i = 1) = \Theta_p(N_s)$), so writing high-dimensional rates in terms of N is equivalent (up to constants) to writing them in terms of the arm-specific sample sizes. Under the balanced-fold assumption above, the estimator can equivalently be written using the combined-fold empirical average as

$$\hat{\mu}_{t,s}^{1,\text{CF}} = \frac{1}{K_f} \sum_{k=1}^{K_f} \tilde{E}_k [\tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)})],$$

which matches the $\tilde{E}_k$ notation used in the remainder decomposition below.

Define the (uncentered) score contribution at a generic nuisance parameter $\vartheta = (\alpha^1, \gamma_1)$:

$$\tilde{\varphi}_{t,s}(D_i; \vartheta) := \frac{I(R_i = t)}{\rho_t} \psi(\phi(\mathbf{X}_i); \alpha^1) + \frac{I(R_i = s)}{\rho_s} w(\mathbf{X}_i; \gamma_1) \{Y_i - \psi(\phi(\mathbf{X}_i); \alpha^1)\}.$$

Define the corresponding population estimand at ϑ as

$$\mu^1(\vartheta) := \mathbb{E}[\tilde{\varphi}_{t,s}(D; \vartheta)] = \mathbb{E}_t[\psi(\phi(\mathbf{X}); \alpha^1)] + \mathbb{E}_s[w(\mathbf{X}; \gamma_1)\{Y - \psi(\phi(\mathbf{X}); \alpha^1)\}].$$

A.3 Common Notation and Definitions

To avoid defining objects inside later statements, we collect here the auxiliary definitions used throughout the appendix.

Definition A.1 (Sub-Gaussian Orlicz norm). *For a random variable Z , the sub-Gaussian Orlicz norm is*

$$\|Z\|_{\psi_2} := \inf\{t > 0 : \mathbb{E}[\exp(Z^2/t^2)] \leq 2\},$$

and Z is called sub-Gaussian when $\|Z\|_{\psi_2} < \infty$.

Definition A.2 (Sparsity indices). *For the population targets of the calibrated losses, define the ℓ_0 -sparsity levels*

$$s_\alpha := \|\alpha^{1,*}\|_0, \quad s_\gamma := \|\gamma_1^*\|_0, \quad s := s_\alpha + s_\gamma,$$

and throughout the appendix we use s (rather than the equivalent notation s_{tot}) for the total sparsity. The same convention applies to the initial nuisance estimators under the two-level alignment of Assumption A.3(c).

Definition A.3 (Target outcome regression). *The target-site treated conditional mean is*

$$m_t^1(\mathbf{X}) := \mathbb{E}_t[Y \mid A = 1, \mathbf{X}],$$

so that under ignorability and overlap on the target site, $\mu^1 = \mathbb{E}_t[m_t^1(\mathbf{X})]$.

Remark A.1 (On $\phi(\mathbf{X})$). *Throughout, $\phi : \mathbb{R}^p \rightarrow \mathbb{R}^p$ denotes a basis expansion of the raw covariates $\mathbf{X}$ that includes an intercept coordinate and is treated as fixed. All high-dimensional and sparsity statements below apply to $\phi(\mathbf{X})$; if the user does not require an explicit expansion, ϕ can be taken as the identity, in which case $p = \dim(\mathbf{X})$.*

Lemma A.1 (Identification / double robustness for $\mu^1(\vartheta^*)$). *Assume the target causal parameter is identified as $\mu^1 = \mathbb{E}_t[Y(1)]$ (e.g., by consistency and ignorability/overlap on the target site), so that $\mu^1 = \mathbb{E}_t[m_t^1(\mathbf{X})]$ where $m_t^1(\mathbf{X}) := \mathbb{E}_t[Y \mid A = 1, \mathbf{X}]$. Recall $\phi(\mathbf{X})$ includes an intercept. Let $\vartheta^* = (\alpha^{1,*}, \gamma_1^*)$ denote the population targets of the calibrated losses. Then $\mu^1(\vartheta^*) = \mu^1$ holds under either of the following conditions.*

1. **Outcome-model correctness.** *If $\psi(\phi(\mathbf{X}); \alpha^{1,*}) = m_t^1(\mathbf{X})$ almost surely, then the calibrated outcome loss implies the population moment condition $\mathbb{E}_s[w(\mathbf{X}; \gamma_1^*) \phi(\mathbf{X}) \{Y - \psi(\phi(\mathbf{X}); \alpha^{1,*})\}] = \mathbf{0}$ (see Lemma A.9(ii) below). In particular, the intercept component yields $\mathbb{E}_s[w(\mathbf{X}; \gamma_1^*) \{Y - \psi(\phi(\mathbf{X}); \alpha^{1,*})\}] = 0$, and therefore $\mu^1(\vartheta^*) = \mathbb{E}_t[\psi(\phi(\mathbf{X}); \alpha^{1,*})] = \mathbb{E}_t[m_t^1(\mathbf{X})] = \mu^1$.*
2. **Weight-model correctness.** *Suppose the weighting model is correct in the sense that for any square-integrable measurable function $h(\mathbf{X})$,*

$$\mathbb{E}_s[w(\mathbf{X}; \gamma_1^*) h(\mathbf{X})] = \mathbb{E}_t[h(\mathbf{X})],$$

and, for the treated potential outcome, the usual covariate-shift/transportability condition holds for this pair (t, s) : $\mathbb{E}_s[Y(1) \mid \mathbf{X}] = \mathbb{E}_t[Y(1) \mid \mathbf{X}]$ almost surely. Then for any outcome model $m(\mathbf{X})$,

$$\mathbb{E}_s[w(\mathbf{X}; \gamma_1^*) \{Y(1) - m(\mathbf{X})\}] = \mathbb{E}_t[Y(1) - m(\mathbf{X})],$$

and hence $\mu^1(\alpha, \gamma_1^*) = \mu^1$ for any α , in particular $\mu^1(\vartheta^*) = \mu^1$ even if the outcome model is misspecified.

Define the centered influence function at the population parameters ϑ^* :

$$\begin{aligned}\varphi_{t,s}(D_i; \vartheta^*) &:= \tilde{\varphi}_{t,s}(D_i; \vartheta^*) - \mu^1(\vartheta^*) \\ &= \frac{I(R_i = t)}{\rho_t} \left(\psi(\phi(\mathbf{X}_i); \boldsymbol{\alpha}^{1,*}) - \mu_t^1(\boldsymbol{\alpha}^{1,*}) \right) \\ &\quad + \frac{I(R_i = s)}{\rho_s} \left(w(\mathbf{X}_i; \boldsymbol{\gamma}_1^*) \{Y_i - \psi(\phi(\mathbf{X}_i); \boldsymbol{\alpha}^{1,*})\} - \delta_s^1(\vartheta^*) \right).\end{aligned}$$

where $\mu_t^1(\boldsymbol{\alpha}^{1,*}) = \mathbb{E}_t[\psi(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})]$ and $\delta_s^1(\vartheta^*) = \mathbb{E}_s[w(\mathbf{X}; \boldsymbol{\gamma}_1^*)\{Y - \psi(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})\}]$. More generally, for any ϑ we define $\varphi_{t,s}(D; \vartheta) := \tilde{\varphi}_{t,s}(D; \vartheta) - \mu^1(\vartheta)$. Let $V_{t,s}^* := \text{Var}(\varphi_{t,s}(D; \vartheta^*))$.

A.4 Assumptions

The assumptions below are organized so that each condition has a clearly identified role: Assumption A.1 controls the design distribution, Assumption A.2 controls the weight tails together with the outcome smoothness and moments, and Assumption A.3 controls the nuisance estimation pipeline. Definitions of Orlicz norms, sparsity, and the target outcome regression are collected in Section A.3.

Assumption A.1 (Design distribution). *The coordinates of $\phi(\mathbf{X})$ (which includes an intercept) are sub-Gaussian in the sense of Definition A.1: there exists $K_\phi < \infty$ such that*

$$\max_{1 \leq j \leq p} \|\phi_j(\mathbf{X})\|_{\psi_2} \leq K_\phi \quad \text{under both } R = t \text{ and } R = s.$$

Assumption A.2 (Overlap, smoothness, and moments). (a) (Bounded-weight overlap) *There exists a constant $\bar{W} < \infty$ such that the ideal calibration weight at the population target is essentially bounded:*

$$w(\mathbf{X}; \boldsymbol{\gamma}_1^*) \leq \bar{W} \quad \text{almost surely under } R = s,$$

equivalently, $\phi(\mathbf{X})^\top \boldsymbol{\gamma}_1^ \geq -\log \bar{W}$ a.s. Under this condition, numerical truncation in the calibrated losses is asymptotically inactive at the target.*

- (b) (GLM smoothness) *The mean function $\psi : \mathbb{R} \rightarrow \mathbb{R}$ is continuously differentiable with $\sup_u |\psi'(u)| \leq L_\psi < \infty$, and ψ' is Lipschitz with constant $L_{\psi'} < \infty$.*
- (c) (Outcome moments) *There exists $\delta > 0$ such that $\mathbb{E}[|Y|^{2+\delta} \mid R = r] < \infty$ for $r \in \{t, s\}$. A sufficient condition, used in the concentration steps below, is that $Y \mid (A = 1, R = r)$ is sub-Gaussian.*
- (d) (Local IF moments) *With the same δ as in (c), there exists a neighborhood $\mathcal{N}(\vartheta^*)$ of the population target such that the uncentered influence-function contribution $\tilde{\varphi}_{t,s}$ defined in the Setup has a uniformly bounded $(2 + \delta)$ -th moment,*

$$\sup_{\vartheta \in \mathcal{N}(\vartheta^*)} \mathbb{E}[|\tilde{\varphi}_{t,s}(D; \vartheta)|^{2+\delta}] < \infty.$$

Under (a), (b), (c) and Assumption A.1, this condition is automatic: the bounded weight and sub-Gaussian $\phi(\mathbf{X})$ imply finite moments of $w(\mathbf{X}; \boldsymbol{\gamma}) \psi(\phi(\mathbf{X})^\top \boldsymbol{\alpha})$ of any order in a neighborhood of ϑ^ .*

- (e) (Inactive truncation) *With $M_{\tau,N}$ as in the definition of $\mathcal{T}(\cdot)$, for $r \in \{t, s\}$,*

$$P\left(|\phi(\mathbf{X})^\top \boldsymbol{\alpha}^{1,*}| > M_{\tau,N} \mid R = r\right) = o(N^{-1}), \quad P\left(|\phi(\mathbf{X})^\top \boldsymbol{\gamma}_1^*| > M_{\tau,N} \mid R = s\right) = o(N^{-1}).$$

This is stronger than strictly necessary but keeps the exposition clean; weakening it to $o(1)$ is possible at the cost of tracking a clipping-induced bias term of order $o_p(N^{-1/2})$.

Assumption A.3 (Nuisance estimation pipeline). (a) (Restricted strong convexity) *The population versions of the calibrated loss functions (Equations (50) and (51) in the main manuscript) are convex and satisfy Restricted Strong Convexity (RSC) around ϑ^* in the relevant sparse norms.*

- (b) (Cross-fitting) *The estimators $\hat{\vartheta}^{(-k)}$ used for evaluation on fold k are trained solely on $\mathcal{D}^{(-k)}$, which is independent of $\mathcal{D}_k$.*

- (c) (*Two-level limit alignment*) The probability limits $(\alpha_{\text{init}}^{1,*}, \gamma_{1,\text{init}}^*)$ of the initial plug-in nuisance estimators coincide with the population targets of the calibrated step, i.e., $\alpha_{\text{init}}^{1,*} = \alpha^{1,*}$ and $\gamma_{1,\text{init}}^* = \gamma_1^*$.
- (d) (*Foldwise score-centering; optional*) The conditional foldwise score-centering condition of the Special case in Lemma A.8 holds for the calibrated losses used to estimate α^1 and γ_1 ; this is the condition exploited by Hou et al. [2025] to eliminate the plug-in score term. When (d) is not verified, the General rate bound in Lemma A.8 applies instead.

Remark A.2 (On the alignment condition). Assumption A.3(c) is a technical device to avoid tracking multiple population limits induced by the nested/coupled optimization. It is satisfied, for example, when (i) the population calibrated losses have unique minimizers $(\alpha^{1,*}, \gamma_1^*)$ (by strict convexity in the relevant restricted set), and (ii) the mapping from an initial plug-in value to the population minimizer of the next calibrated update is continuous at $(\alpha^{1,*}, \gamma_1^*)$, so that any consistent initialization converges to the same fixed point. In practice, the same calibrated estimating equations are solved throughout the nuisance-estimation pipeline, making the alignment natural under uniqueness.

A.5 Oracle Rates for the Calibrated Nuisance Estimators

Before stating the asymptotic normality of the cross-fitted estimator, we record the high-dimensional oracle rates of the calibrated nuisance estimators. The result is a direct specialization of the cross-fitted ℓ_1 -penalized nuisance rate bound of Lemma A.8 (which in turn adapts Lemma A21 of Hou et al. [2025] and the RSC framework of Negahban et al. [2012], Tan [2020], Bühlmann and Van De Geer [2011]) to the calibrated outcome and calibration-weight losses of Equations (51) and (50). Separating this oracle step from the subsequent asymptotic-normality argument keeps the role of the nuisance rates logically distinct from the debiasing argument and facilitates reuse across the pairwise and aggregated statements.

Proposition A.2 (Oracle rates of calibrated nuisance estimators). *Under Assumptions A.1, A.2, and A.3(a)–(c), with regularization parameters $\lambda_\alpha \asymp \sqrt{\log(p)/N}$ and $\lambda_\gamma \asymp \sqrt{\log(p)/N}$, the calibrated cross-fitted ℓ_1 -penalized nuisance estimators satisfy*

$$\|\hat{\alpha}_{t,s}^{1,(-k)} - \alpha^{1,*}\|_2 = O_p\left(\sqrt{\frac{s_\alpha \log(p)}{N}}\right), \quad \|\hat{\alpha}_{t,s}^{1,(-k)} - \alpha^{1,*}\|_1 = O_p\left(s_\alpha \sqrt{\frac{\log(p)}{N}}\right),$$

and analogously for $\hat{\gamma}_{s,1}^{(-k)}$ with s_α replaced by s_γ . If Assumption A.3(d) additionally holds, the foldwise score-centering is available and the Special rate bound in Lemma A.8 applies without inflation from the plug-in score term.

Proof. The proposition is obtained by applying Lemma A.8 twice, once to the calibrated outcome loss (Equation (51) in the main manuscript) and once to the calibrated calibration-weight loss (Equation (50)). We verify each condition of Lemma A.8 and then derive the stated rates.

Step 1 (Outcome loss as an instance of Lemma A.8). Identify the generic estimation target of Lemma A.8 with $\beta \equiv \alpha^1$, the plug-in nuisance with $\gamma \equiv \gamma_1$, and the per-observation loss with

$$\ell(D; \alpha^1, \gamma_1) = \frac{I(R=s)}{\rho_s} \frac{I(A=1)}{e^{\mathcal{T}(\phi(\mathbf{X})^\top \gamma_1)}} \left[-Y \phi(\mathbf{X})^\top \alpha^1 + \Psi(\phi(\mathbf{X})^\top \alpha^1) \right].$$

We check the conditions (a)–(d) of Lemma A.8:

- (a) Sub-Gaussian design with bounded ψ_2 -norm is supplied by Assumption A.1.
- (b) Response sub-Gaussianity is Assumption A.2(c); the GLM mean function ψ and its derivative ψ' are uniformly bounded on a neighborhood of $\alpha^{1,*}$ by Assumption A.2(b), using that $\phi(\mathbf{X})^\top \alpha^{1,*}$ has a sub-Gaussian distribution by Assumption A.1.
- (c) Identifiability of the population Gram matrix $\mathbb{E}[\phi(\mathbf{X})\phi(\mathbf{X})^\top \mid R=s]$ follows from the RSC of the outcome calibrated loss (Assumption A.3(a)): RSC with curvature $\kappa_\alpha > 0$ implies that the restricted minimum eigenvalue of the population Hessian at $\alpha^{1,*}$ is bounded below, which, combined with bounded weights (Assumption A.2(a)), yields a strictly positive minimum eigenvalue of the unrestricted Gram matrix on the support of $\alpha^{1,*}$ and in its local extensions.
- (d) Weight regularity: the calibration weight $w(\mathbf{X}; \gamma)$ is locally Lipschitz in γ because $u \mapsto e^{-u}$ is smooth, and its moments are finite under (a) of Assumption A.2.

Cross-fitting independence is given by Assumption A.3(b), and the alignment condition (c) guarantees that the initial plug-in $\hat{\gamma}_1^{(-k)}$ converges to the same population target γ_1^* used in the analysis. With $\lambda_\alpha \asymp \sqrt{\log(p)/N}$, Lemma A.8 yields the stated ℓ_2 and ℓ_1 rates for $\hat{\alpha}_{t,s}^{1,(-k)} - \alpha^{1,*}$ with the sparsity index s_α of Definition A.2.

Step 2 (Calibration-weight loss). Identify the generic target with $\beta \equiv \gamma_1$, the plug-in nuisance with $\gamma \equiv \alpha^1$, and the per-observation loss with

$$\ell(D; \gamma_1, \alpha^1) = \frac{I(R=t)}{\rho_t} \psi'(\mathcal{T}(\phi(\mathbf{X})^\top \alpha^1)) \phi(\mathbf{X})^\top \gamma_1 + \frac{I(R=s)}{\rho_s} \frac{I(A=1) \psi'(\mathcal{T}(\phi(\mathbf{X})^\top \alpha^1))}{e^{\phi(\mathbf{X})^\top \gamma_1}}.$$

The loss is strictly convex in γ_1 because $u \mapsto e^{-u}$ is strictly convex, and its Hessian at γ_1^* equals $\mathbb{E}_s[w(\mathbf{X}; \gamma_1^*) \psi'(\phi(\mathbf{X})^\top \alpha^{1,*}) \phi(\mathbf{X}) \phi(\mathbf{X})^\top] / \rho_s$, which is positive definite on the support of γ_1^* by Assumption A.3(a). Conditions (a)–(d) of Lemma A.8 are verified exactly as in Step 1 (with α^1 and γ_1 swapped), using that $\sup_u |\psi'(u)| \leq L_\psi$ (Assumption A.2(b)) and the bounded-weight condition (Assumption A.2(a)). With $\lambda_\gamma \asymp \sqrt{\log(p)/N}$, Lemma A.8 delivers the stated rates for $\hat{\gamma}_{s,1}^{(-k)} - \gamma_1^*$ with sparsity index s_γ .

Step 3 (Foldwise score-centering). If Assumption A.3(d) additionally holds, the foldwise conditional score-centering condition required by the *Special* branch of Lemma A.8 is satisfied for both calibrated losses, and the corresponding rate bound applies without the plug-in score term; otherwise, the *General* branch of the lemma applies, giving the same stated rates up to constants. $\square$

The asymptotic-normality statements in Sections A.6 and B will invoke Proposition A.2 exclusively through the product rate

$$\|\hat{\alpha}_{t,s}^{1,(-k)} - \alpha^{1,*}\|_2 \cdot \|\hat{\gamma}_{s,1}^{(-k)} - \gamma_1^*\|_2 = O_p\left(\frac{\sqrt{s_\alpha s_\gamma \log(p)}}{N}\right),$$

which, combined with the cross-calibration property of Lemma A.9, drives the bias remainder $R_{2,k}$ to $o_p(N^{-1/2})$.

A.6 Main Theorem and Proof

Theorem A.3 (Asymptotic Normality of $\hat{\mu}_{t,s}^{1,\text{CF}}$). *Suppose Assumptions A.1, A.2, and A.3(a)–A.3(c) hold. (Assumption A.3(d) is an optional strengthening used only to invoke the **Special** rate in Lemma A.8; the theorem also holds under the **General** rate condition there.) Assume ϑ^* is the unique population target pair induced by the calibrated losses under the alignment condition in Assumption A.3(c), implying Lemma A.9 holds. If further, at least one nuisance model (outcome or weighting) is correctly specified in the sense of Lemma A.1 (so $\mu^1(\vartheta^*) = \mu^1$), and the total sparsity $s = s_\alpha + s_\gamma$ of Definition A.2 (applied to both the initial and the calibrated nuisance targets under the alignment condition of Assumption A.3(c)) satisfies*

$$\frac{s^2 \log^2(p)}{N} \rightarrow 0,$$

then the cross-fitted estimator is asymptotically normal:

$$\sqrt{N} (\hat{\mu}_{t,s}^{1,\text{CF}} - \mu^1) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^*).$$

Moreover, the finite-sample KKT perturbation from the ℓ_1 penalties is absorbed in the remainder term (via Lemma A.8) and is $o_p(N^{-1/2})$ under the stated sparsity/rate condition.

Proof. Suppose in addition that at least one nuisance model is correctly specified in the sense of Lemma A.1, so that $\mu^1(\vartheta^*) = \mu^1$. We may then center the expansion at μ^1 . The total estimation error is decomposed as: Write the decomposition

$$\hat{\mu}_{t,s}^{1,\text{CF}} - \mu^1 = \underbrace{\left(\frac{1}{N} \sum_{i=1}^N \varphi_{t,s}(D_i; \vartheta^*) \right)}_{\mathbb{T}_I} + \underbrace{\left(\hat{\mu}_{t,s}^{1,\text{CF}} - \frac{1}{N} \sum_{i=1}^N \varphi_{t,s}(D_i; \vartheta^*) - \mu^1 \right)}_{\mathbb{T}_{II}},$$

where $\mathbf{T}_I$ is the average of centered influence functions at the population target and $\mathbf{T}_{II}$ is the remainder; $\mathbb{E}[\varphi_{t,s}(D; \vartheta^*)] = 0$ by construction and $\mu^1(\vartheta^*) = \mu_t^1(\boldsymbol{\alpha}^{1,*}) + \delta_s^1(\vartheta^*)$. By Lemma A.5,

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N \varphi_{t,s}(D_i; \vartheta^*) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^*),$$

so $\sqrt{N} \mathbf{T}_I \xrightarrow{d} \mathcal{N}(0, V_{t,s}^*)$. The remainder is an average over the folds, $\mathbf{T}_{II} = \frac{1}{K_f} \sum_{k=1}^{K_f} (R_{1,k} + R_{2,k})$, with components analyzed below. $R_{1,k}$ is the conditional empirical process fluctuation of the *difference* score:

$$R_{1,k} = \left(\tilde{E}_k - \mathbb{E}[\cdot \mid \mathcal{D}^{(-k)}] \right) \left[\tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)}) - \tilde{\varphi}_{t,s}(D; \vartheta^*) \right].$$

By cross-fitting, conditional on $\mathcal{D}^{(-k)}$, the summands are independent (with a two-sample/triangular-array structure) on fold k with mean zero. Using Lemma A.10 (Part 2),

$$\mathbb{E} \left[\left(\tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)}) - \tilde{\varphi}_{t,s}(D; \vartheta^*) \right)^2 \mid \mathcal{D}^{(-k)} \right] \leq C \left(\|\hat{\boldsymbol{\alpha}}_{t,s}^{1,(-k)} - \boldsymbol{\alpha}^{1,*}\|_2^2 + \|\hat{\boldsymbol{\gamma}}_{s,1}^{(-k)} - \boldsymbol{\gamma}_1^*\|_2^2 \right),$$

and therefore the conditional variance of $R_{1,k}$ obeys

$$\text{Var}(R_{1,k} \mid \mathcal{D}^{(-k)}) = \text{Var} \left(\tilde{E}_k[\Delta_k(D)] \mid \mathcal{D}^{(-k)} \right) \leq \frac{1}{N_k} \mathbb{E}[\Delta_k(D)^2 \mid \mathcal{D}^{(-k)}],$$

where $\Delta_k(D) := \tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)}) - \tilde{\varphi}_{t,s}(D; \vartheta^*)$. Here we used that, conditional on $\mathcal{D}^{(-k)}$, the $\{\Delta_k(D_i) : i \in \mathcal{D}_k\}$ are independent, so

$$\begin{aligned} \text{Var} \left(\tilde{E}_k[\Delta_k(D)] \mid \mathcal{D}^{(-k)} \right) &= \text{Var} \left(\tilde{E}_k[\Delta_k(D)] - \mathbb{E}[\Delta_k(D) \mid \mathcal{D}^{(-k)}] \mid \mathcal{D}^{(-k)} \right) \\ &\leq \frac{1}{N_k} \mathbb{E}[(\Delta_k(D) - \mathbb{E}[\Delta_k(D) \mid \mathcal{D}^{(-k)}])^2 \mid \mathcal{D}^{(-k)}] \\ &\leq \frac{1}{N_k} \mathbb{E}[\Delta_k(D)^2 \mid \mathcal{D}^{(-k)}]. \end{aligned}$$

Since $N_k \asymp N$ for fixed K_f , Chebyshev's inequality yields

$$|R_{1,k}| = O_p \left(\frac{\|\hat{\boldsymbol{\alpha}}_{t,s}^{1,(-k)} - \boldsymbol{\alpha}^{1,*}\|_2 + \|\hat{\boldsymbol{\gamma}}_{s,1}^{(-k)} - \boldsymbol{\gamma}_1^*\|_2}{\sqrt{N}} \right) = O_p \left(\frac{\sqrt{s \log(p)}}{N} \right) = o_p(N^{-1/2}).$$

$R_{2,k}$ is the first-order bias remainder, conditional on $\mathcal{D}^{(-k)}$:

$$R_{2,k} = \mathbb{E}[\tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)}) - \tilde{\varphi}_{t,s}(D; \vartheta^*) \mid \mathcal{D}^{(-k)}]$$

By Lemma A.10 (Part 1), which relies on the orthogonality conditions derived in Lemma A.9, this bias is bounded by a second-order term:

$$|R_{2,k}| \leq C \cdot \left(\|\hat{\boldsymbol{\alpha}}_{t,s}^{1,(-k)} - \boldsymbol{\alpha}^{1,*}\|_2^2 + \|\hat{\boldsymbol{\gamma}}_{s,1}^{(-k)} - \boldsymbol{\gamma}_1^*\|_2^2 + \|\hat{\boldsymbol{\alpha}}_{t,s}^{1,(-k)} - \boldsymbol{\alpha}^{1,*}\|_2 \|\hat{\boldsymbol{\gamma}}_{s,1}^{(-k)} - \boldsymbol{\gamma}_1^*\|_2 \right)$$

Using the nuisance rates from Lemma A.8:

$$|R_{2,k}| = O_p \left(\frac{s_\alpha \log(p)}{N} \right) + O_p \left(\frac{s_\gamma \log(p)}{N} \right) + O_p \left(\frac{\sqrt{s_\alpha s_\gamma \log(p)}}{N} \right) = O_p \left(\frac{s \log(p)}{N} \right).$$

Averaging over k gives the total bias remainder:

$$R_2 = \frac{1}{K_f} \sum_{k=1}^{K_f} R_{2,k} = O_p \left(\frac{s \log(p)}{N} \right).$$

We require $\sqrt{N} \cdot R_2 \rightarrow 0$, i.e., $O_p(s \log(p)/\sqrt{N}) \rightarrow 0$, which is implied by the theorem's condition $s^2 \log^2(p)/N \rightarrow 0$. Thus the total remainder $\mathbf{T}_{II} = R_1 + R_2$ satisfies $\sqrt{N} \mathbf{T}_{II} = o_p(1)$, and hence

$$\sqrt{N} (\hat{\mu}_{t,s}^{1,\text{CF}} - \mu^1) = \frac{1}{\sqrt{N}} \sum_{i=1}^N \varphi_{t,s}(D_i; \vartheta^*) + o_p(1).$$

Together with Lemma A.5, this yields $\sqrt{N} (\hat{\mu}_{t,s}^{1,\text{CF}} - \mu^1) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^*)$ by Slutsky's theorem. $\square$

Lemma A.4 (Consistent variance estimation). *Assume Assumptions A.1–A.3 and that the population asymptotic variance is bounded away from zero, $V_{t,s}^* \geq \underline{v} > 0$. Then the plug-in estimator based on the cross-fitted empirical influence function is consistent:*

$$\widehat{V}_{t,s} = \frac{1}{K_f} \sum_{k=1}^{K_f} \widetilde{E}_k \left[\left(\widetilde{\varphi}_{t,s}(D; \widehat{\vartheta}^{(-k)}) - \widehat{\mu}_{t,s}^{1,\text{CF}} \right)^2 \right] \xrightarrow{p} V_{t,s}^*.$$

Proof. The argument combines three ingredients: the nuisance consistency rates of Proposition A.2 (equivalently, $\|\widehat{\vartheta}^{(-k)} - \vartheta^*\|_2 = o_p(1)$), the local L_2 -Lipschitz continuity of $\vartheta \mapsto \widetilde{\varphi}_{t,s}(D; \vartheta)$ (there exists $C < \infty$ such that, for ϑ_1, ϑ_2 in a neighborhood of ϑ^* , $\mathbb{E}[(\widetilde{\varphi}_{t,s}(D; \vartheta_1) - \widetilde{\varphi}_{t,s}(D; \vartheta_2))^2] \leq C \|\vartheta_1 - \vartheta_2\|_2^2$, which follows from Assumptions A.2(a)–(b) as in Lemma A.10(Part 2)), and uniform integrability of $\{\widetilde{\varphi}_{t,s}(D; \widehat{\vartheta}^{(-k)})^2\}_{k,N}$ (supplied by the $(2 + \delta)$ -moment bound in Assumption A.2(d)).

Step 1 (Reduce to within-fold centering). For each fold k , define the foldwise score

$$Z_i^{(k)} := \widetilde{\varphi}_{t,s}(D_i; \widehat{\vartheta}^{(-k)}), \quad i \in \mathcal{D}_k,$$

and the foldwise mean $\widehat{\mu}_k := \widetilde{E}_k[Z^{(k)}]$. Since $\widehat{\mu}_k^{1,\text{CF}} = \frac{1}{K_f} \sum_{k=1}^{K_f} \widehat{\mu}_k$, we have the identity

$$\widetilde{E}_k \left[(Z^{(k)} - \widehat{\mu}_k^{1,\text{CF}})^2 \right] = \widetilde{E}_k \left[(Z^{(k)} - \widehat{\mu}_k)^2 \right] + \left(\widehat{\mu}_k - \widehat{\mu}_k^{1,\text{CF}} \right)^2,$$

where the cross term vanishes because $\widetilde{E}_k[Z^{(k)} - \widehat{\mu}_k] = 0$ by definition of $\widehat{\mu}_k$. Therefore,

$$\widehat{V}_{t,s} = \frac{1}{K_f} \sum_{k=1}^{K_f} \widetilde{E}_k \left[(Z^{(k)} - \widehat{\mu}_k)^2 \right] + \frac{1}{K_f} \sum_{k=1}^{K_f} \left(\widehat{\mu}_k - \widehat{\mu}_k^{1,\text{CF}} \right)^2.$$

By Lemma A.8, $\widehat{\vartheta}^{(-k)} = \vartheta^* + o_p(1)$, and by the conditional LLN below we have $\widehat{\mu}_k = \mu^1(\vartheta^*) + o_p(1)$ for each fixed k . Since K_f is fixed, this implies $\max_{1 \leq k \leq K_f} \left| \widehat{\mu}_k - \widehat{\mu}_k^{1,\text{CF}} \right| = o_p(1)$, and hence the second term is $o_p(1)$. Thus it suffices to show

$$\frac{1}{K_f} \sum_{k=1}^{K_f} \widetilde{E}_k \left[(Z^{(k)} - \widehat{\mu}_k)^2 \right] \xrightarrow{p} V_{t,s}^*.$$

Step 2 (Conditional LLN for first and second moments). Fix a fold k . Conditional on the training sample $\mathcal{D}^{(-k)}$, the nuisance estimate $\widehat{\vartheta}^{(-k)}$ is fixed and independent of $\mathcal{D}_k$. Therefore, conditional on $\mathcal{D}^{(-k)}$, the random variables $\{Z_i^{(k)} : i \in \mathcal{D}_k\}$ are independent with finite $(2 + \delta)$ moments uniformly in N by Assumption A.2(b). By the (conditional) weak law of large numbers,

$$\widehat{\mu}_k - \mathbb{E}[Z^{(k)} \mid \mathcal{D}^{(-k)}] = o_p(1), \quad \widetilde{E}_k[(Z^{(k)})^2] - \mathbb{E}[(Z^{(k)})^2 \mid \mathcal{D}^{(-k)}] = o_p(1).$$

Since $\widetilde{E}_k[(Z^{(k)} - \widehat{\mu}_k)^2] = \widetilde{E}_k[(Z^{(k)})^2] - \widehat{\mu}_k^2$, combining the two displays yields

$$\widetilde{E}_k[(Z^{(k)} - \widehat{\mu}_k)^2] - \left\{ \mathbb{E}[(Z^{(k)})^2 \mid \mathcal{D}^{(-k)}] - \mathbb{E}[Z^{(k)} \mid \mathcal{D}^{(-k)}]^2 \right\} = o_p(1).$$

Step 3 (Plug-in stability: conditional expectations converge to population targets). It remains to show that the bracketed conditional variance converges in probability to $V_{t,s}^*$. First, by nuisance consistency (Ingredient 1) and the L_2 -Lipschitz property (Ingredient 2),

$$\mathbb{E} \left[\left(Z^{(k)} - \widetilde{\varphi}_{t,s}(D; \vartheta^*) \right)^2 \right] = \mathbb{E} \left[\left(\widetilde{\varphi}_{t,s}(D; \widehat{\vartheta}^{(-k)}) - \widetilde{\varphi}_{t,s}(D; \vartheta^*) \right)^2 \right] \leq C \|\widehat{\vartheta}^{(-k)} - \vartheta^*\|_2^2 \xrightarrow{p} 0.$$

Next we establish convergence of the *conditional* first and second moments. Let

$$m_k := \mathbb{E}[Z^{(k)} \mid \mathcal{D}^{(-k)}], \quad q_k := \mathbb{E}[(Z^{(k)})^2 \mid \mathcal{D}^{(-k)}], \quad \mu^* := \mu^1(\vartheta^*) = \mathbb{E}[\widetilde{\varphi}_{t,s}(D; \vartheta^*)].$$

Then

$$m_k - \mu^* = \mathbb{E}[\widetilde{\varphi}_{t,s}(D; \widehat{\vartheta}^{(-k)}) - \widetilde{\varphi}_{t,s}(D; \vartheta^*) \mid \mathcal{D}^{(-k)}],$$

so by conditional Jensen,

$$(m_k - \mu^*)^2 \leq \mathbb{E} \left[\left(\tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)}) - \tilde{\varphi}_{t,s}(D; \vartheta^*) \right)^2 \mid \mathcal{D}^{(-k)} \right].$$

Taking expectations and using the L_2 -Lipschitz bound in Ingredient 2 gives $\mathbb{E}[(m_k - \mu^*)^2] \rightarrow 0$, hence $m_k \rightarrow_p \mu^*$. Similarly, write

$$q_k - \mathbb{E}[\tilde{\varphi}_{t,s}(D; \vartheta^*)^2] = \mathbb{E} \left[(Z^{(k)})^2 - \tilde{\varphi}_{t,s}(D; \vartheta^*)^2 \mid \mathcal{D}^{(-k)} \right].$$

Using $(a^2 - b^2) = (a - b)(a + b)$ and conditional Cauchy–Schwarz,

$$\begin{aligned} |q_k - \mathbb{E}[\tilde{\varphi}_{t,s}(D; \vartheta^*)^2]| &= \left| \mathbb{E} \left[(Z^{(k)})^2 - \tilde{\varphi}_{t,s}(D; \vartheta^*)^2 \mid \mathcal{D}^{(-k)} \right] \right| \\ &\leq \mathbb{E} \left[\left| Z^{(k)} - \tilde{\varphi}_{t,s}(D; \vartheta^*) \right| \cdot \left| Z^{(k)} + \tilde{\varphi}_{t,s}(D; \vartheta^*) \right| \mid \mathcal{D}^{(-k)} \right] \\ &\leq \left(\mathbb{E} \left[\left(Z^{(k)} - \tilde{\varphi}_{t,s}(D; \vartheta^*) \right)^2 \mid \mathcal{D}^{(-k)} \right] \right)^{1/2} \left(\mathbb{E} \left[\left(Z^{(k)} + \tilde{\varphi}_{t,s}(D; \vartheta^*) \right)^2 \mid \mathcal{D}^{(-k)} \right] \right)^{1/2}. \end{aligned}$$

The first factor converges to 0 in probability by the L_2 -Lipschitz bound in Ingredient 2 and nuisance consistency (Ingredient 1). The second factor is $O_p(1)$ by the $(2 + \delta)$ moment assumption in Assumption A.2(b) (uniform integrability). Therefore $q_k \rightarrow_p \mathbb{E}[\tilde{\varphi}_{t,s}(D; \vartheta^*)^2]$. Consequently, since $m_k \rightarrow_p \mu^*$,

$$\mathbb{E} \left[(Z^{(k)})^2 \mid \mathcal{D}^{(-k)} \right] - \mathbb{E} \left[Z^{(k)} \mid \mathcal{D}^{(-k)} \right]^2 \xrightarrow{p} \mathbb{E}[\tilde{\varphi}_{t,s}(D; \vartheta^*)^2] - \mu^1(\vartheta^*)^2 = \text{Var}(\tilde{\varphi}_{t,s}(D; \vartheta^*)) = V_{t,s}^*.$$

Step 4 (Average over folds). Since K_f is fixed, combining Step 2 and Step 3 and averaging over k yields $\hat{V}_{t,s} \xrightarrow{p} V_{t,s}^*$.

Extension to the aggregated estimator. For the weighted aggregated estimator with inner M -fold cross-fitted weights $\hat{\eta}^{(-k)}$, we define per-observation *uncentered* scaled influence functions that serve as the unified estimating equation for both the point estimate and the variance.

Unified estimating equation. For each observation i evaluated on fold $\mathcal{D}_{k(i)}$, define the uncentered IF

$$\hat{\Phi}_i := \begin{cases} \frac{N_{\text{all}}}{N_t} \left[(1 - \sum_j \hat{\eta}_j^{(-k(i))}) \tilde{Y}_{\text{ot},i} + \sum_j \hat{\eta}_j^{(-k(i))} \psi(\phi(\mathbf{X}_i); \hat{\alpha}_{t,s_j}^{1,(-k(i))}) \right], & i \in \mathcal{D}_t, \\ \frac{N_{\text{all}}}{N_{s_j}} \hat{\eta}_j^{(-k(i))} w(\mathbf{X}_i; \hat{\gamma}_{s_j,1}^{(-k(i))}) (Y_i - \psi(\phi(\mathbf{X}_i); \hat{\alpha}_{t,s_j}^{1,(-k(i))})), & i \in \mathcal{D}_{s_j}, \end{cases}$$

Here $\tilde{Y}_{\text{ot},i}$ is the *uncentered* AIPW pseudo-outcome for the target-only estimator,

$$\tilde{Y}_{\text{ot},i} := \frac{A_i Y_i}{\hat{\pi}_{\text{ot}}(\mathbf{X}_i)} - \left(\frac{A_i}{\hat{\pi}_{\text{ot}}(\mathbf{X}_i)} - 1 \right) \psi(\phi(\mathbf{X}_i); \hat{\alpha}_{\text{ot}}^{1,(-k(i))}),$$

so that the centered IF satisfies $\hat{\varphi}_{\text{ot},i} = \tilde{Y}_{\text{ot},i} - \hat{\mu}_{\text{ot},k(i)}^1$. Likewise, $\hat{\zeta}_{j,i} = \psi(\phi(\mathbf{X}_i); \hat{\alpha}_{t,s_j}^{1,(-k(i))}) - \tilde{E}_{t,k(i)}[\psi(\cdot)]$. All nuisance parameters $\hat{\alpha}_{t,s_j}^{1,(-k)}$, $\hat{\gamma}_{s_j,1}^{(-k)}$, and $\hat{\pi}_{\text{ot}}^{(-k)}$ are trained on data excluding the evaluation fold $\mathcal{D}_k$ (with the inner M -fold calibration/averaging described above), and $\hat{\eta}^{(-k)}$ is estimated via the inner M -fold loop within $\mathcal{T}_k = \mathcal{D} \setminus \mathcal{D}_k$. Note that the source contribution is written in uncentered form; subtracting the fold-specific correction $\hat{\delta}_{t,s_j}^{(-k)} = \tilde{E}_{s_j,k}[w(Y - \psi)]$ yields the centered influence component $\hat{\xi}_{j,i}$ used in the variance-component derivations above.

The point estimate and variance are then derived from the *same* set $\{\hat{\Phi}_i\}_{i=1}^{N_{\text{all}}}$:

$$\begin{aligned} \hat{\mu}_{\text{agg}}^1 &= \bar{\Phi} := \frac{1}{N_{\text{all}}} \sum_{i=1}^{N_{\text{all}}} \hat{\Phi}_i, \\ \hat{V}_{\text{agg}} &= \frac{1}{N_{\text{all}}^2} \sum_{g \in \{t, s_1, \dots, s_K\}} \sum_{i \in \mathcal{D}_g} (\hat{\Phi}_i - \bar{\Phi}_g)^2, \end{aligned}$$

where $\bar{\Phi}_g := \frac{1}{N_g} \sum_{i \in \mathcal{D}_g} \hat{\Phi}_i$ is the *within-group* mean for group $g \in \{t, s_1, \dots, s_K\}$. In the ANOVA decomposition $\text{TSS} = \text{WSS} + \text{BSS}$, the estimator $\hat{V}_{\text{agg}}$ uses only the within-group sum of squares $\text{WSS}/N_{\text{all}}^2$; the between-group sum of squares $\text{BSS}/N_{\text{all}}^2$ is excluded.

Why within-group centering is necessary. Because $\hat{\Phi}_i$ is scaled by N_{all}/N_g (i.e., N_{all}/N_t for target, N_{all}/N_{s_j} for source), the conditional expectations differ across groups: $\mathbb{E}[\hat{\Phi}_i \mid i \in \mathcal{D}_t] \approx (N_{\text{all}}/N_t)\mu^1$ and $\mathbb{E}[\hat{\Phi}_i \mid i \in \mathcal{D}_{s_j}] \approx 0$ (when $\delta_{t,s_j}^* = 0$). The inferential target here is

$$\frac{1}{N_{\text{all}}^2} \sum_{i=1}^{N_{\text{all}}} \text{Var}(\Phi_i),$$

rather than $\text{Var}(\Phi)$ under a pooled-mixture distribution. Global centering at $\bar{\Phi}$ estimates the latter object and introduces an additional between-group term

$$\sum_g \rho_g (\mu_g - \mu)^2,$$

equivalently the ANOVA component $\text{BSS}/N_{\text{all}}^2$, which is of the same order as $V_{\text{agg}}^*/N_{\text{all}}$ and therefore does not vanish. By contrast, $\text{WSS}/N_{\text{all}}^2$ expands as

$$\hat{V}_{\text{agg}} = \frac{1}{N_t} \hat{\sigma}_T^2 + \sum_{j=1}^K \frac{1}{N_{s_j}} \hat{\eta}_j^2 \hat{\sigma}_{S_j}^2,$$

where

$$\hat{\sigma}_T^2 = \widehat{\text{Var}}_t \left((1 - \sum_j \hat{\eta}_j) \hat{\varphi}_{\text{ot},i} + \sum_j \hat{\eta}_j \hat{\zeta}_{j,i} \right).$$

Hence $\hat{\sigma}_T^2$ is not a target-only variance term; it already contains the shared-target cross-covariances (including terms corresponding to $\text{Cov}(\hat{\varphi}_{\text{ot}}, \hat{\zeta}_j)$ and $\text{Cov}(\hat{\zeta}_j, \hat{\zeta}_k)$ for $j \neq k$), fully consistent with the $\hat{C}_{\text{ot},s_j}$ and $\hat{C}_{t,s_j,s_k}$ terms in Equation (22).

With balanced folds ($|\mathcal{D}_{t,k}| = N_t/K_f$, $|\mathcal{D}_{s_j,k}| = N_{s_j}/K_f$), $\bar{\Phi}$ coincides with the fold-average $(1/K_f) \sum_k \hat{\mu}_{\text{agg},k}^1$; with minor fold imbalance the difference is $O(N_{\text{all}}^{-1})$.

This unified form ensures $\hat{\mu}_{\text{agg}}^1 - \mu^1 = \frac{1}{N_{\text{all}}} \sum_{i=1}^{N_{\text{all}}} (\hat{\Phi}_i - \mu^1)$, so the DML representation of the estimation error as an IF average is preserved. By the same arguments as above (nuisance consistency, L_2 -Lipschitz continuity, and $\hat{\eta}^{(-k)} = \eta^* + o_p(1)$), $\hat{V}_{\text{agg}} \xrightarrow{p} V_{\text{agg}}^*$. $\square$

A.7 Technical Lemmas

Lemma A.5 (CLT for IF Average). *Under Assumptions A.1 and A.2, the influence function average is asymptotically normal:*

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N \varphi_{t,s}(D_i; \vartheta^*) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^*).$$

Proof. The observations consist of two independent samples: $\{D_i : i \in \mathcal{D}_t\}$ drawn i.i.d. from the target distribution and $\{D_i : i \in \mathcal{D}_s\}$ drawn i.i.d. from the source distribution, with $N_t/N \rightarrow \rho_t \in (0, 1)$ and $N_s/N \rightarrow \rho_s$. Thus the summands $\{\varphi_{t,s}(D_i; \vartheta^*)\}_{i=1}^N$ are independent (but not identically distributed), have mean zero, and have finite $(2 + \delta)$ moments by Assumption A.2(b). In particular, the Lindeberg condition holds. We verify a Lyapunov condition, which implies the Lindeberg condition for this independent triangular array. Let $X_{i,N} := \varphi_{t,s}(D_i; \vartheta^*)$ so that $\mathbb{E}[X_{i,N}] = 0$ and $\sum_{i=1}^N \text{Var}(X_{i,N}) = NV_{t,s}^* + o(N)$. By Assumption A.2(b), $\sup_{i,N} \mathbb{E}[|X_{i,N}|^{2+\delta}] < \infty$ for some $\delta > 0$. Therefore,

$$\frac{1}{(NV_{t,s}^*)^{1+\delta/2}} \sum_{i=1}^N \mathbb{E}[|X_{i,N}|^{2+\delta}] \leq \frac{CN}{N^{1+\delta/2}} = CN^{-\delta/2} \rightarrow 0,$$

which is the Lyapunov condition. Hence the Lindeberg–Feller CLT yields $\frac{1}{\sqrt{N}} \sum_{i=1}^N X_{i,N} \xrightarrow{d} \mathcal{N}(0, V_{t,s}^*)$, proving the lemma. $\square$

A.8 Extension to the Control Mean and the ATE

The results above were stated for the treated potential outcome mean $\mu^1 = \mathbb{E}_t[Y(1)]$ for notational simplicity. The same proof strategy applies to the control mean

$$\mu^0 := \mathbb{E}_t[Y(0)],$$

by replacing the treatment level 1 by 0 throughout (including the corresponding nuisance parameters (α^0, γ_0) and weights). In particular, define the control-arm weight

$$w^0(\mathbf{X}; \gamma_0) := \frac{I(A=0)}{e^{\phi(\mathbf{X})^T \gamma_0}},$$

and the cross-fitted estimator

$$\hat{\mu}_{t,s}^{0,\text{CF}} = \frac{1}{K_f} \sum_{k=1}^{K_f} \left\{ \tilde{E}_{t,k}[\psi(\phi(\mathbf{X}); \hat{\alpha}_{t,s}^{0,(-k)})] + \tilde{E}_{s,k} \left[w^0(\mathbf{X}; \hat{\gamma}_{s,0}^{(-k)}) (Y - \psi(\phi(\mathbf{X}); \hat{\alpha}_{t,s}^{0,(-k)})) \right] \right\}.$$

Let $\vartheta^{0,*} = (\alpha^{0,*}, \gamma_0^*)$ denote the population targets of the control-arm calibrated losses (the same construction as for μ^1 but with $I(A=0)$), and define the corresponding centered influence function $\varphi_{t,s}^0(D; \vartheta^{0,*})$ analogously.

Theorem A.6 (Asymptotic Normality of $\hat{\mu}_{t,s}^{0,\text{CF}}$). *Under the analogues of Assumptions A.1–A.3 for the control arm, and assuming identification of $\mu^0(\vartheta^{0,*}) = \mu^0$ under outcome-model or weight-model correctness (control-arm analogue of Lemma A.1), if the total sparsity satisfies $\frac{s_{\text{tot}}^2 \log^2(p)}{N} \rightarrow 0$, then*

$$\sqrt{N} (\hat{\mu}_{t,s}^{0,\text{CF}} - \mu^0) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^{0,*}), \quad V_{t,s}^{0,*} := \text{Var}(\varphi_{t,s}^0(D; \vartheta^{0,*})).$$

Proof. The proof is identical to that of Theorem A.3 after replacing the treatment level 1 by 0 in the definition of the score contribution, the calibrated moment conditions, and the corresponding nuisance estimators. $\square$

Corollary A.7 (Asymptotic Normality of the ATE). *Let $\tau := \mu^1 - \mu^0$ and define the cross-fitted ATE estimator*

$$\hat{\tau}_{t,s}^{\text{CF}} := \hat{\mu}_{t,s}^{1,\text{CF}} - \hat{\mu}_{t,s}^{0,\text{CF}}.$$

Under the conditions of Theorems A.3 and A.6, we have the asymptotic linear representation

$$\sqrt{N} (\hat{\tau}_{t,s}^{\text{CF}} - \tau) = \frac{1}{\sqrt{N}} \sum_{i=1}^N \left(\varphi_{t,s}(D_i; \vartheta^*) - \varphi_{t,s}^0(D_i; \vartheta^{0,*}) \right) + o_p(1),$$

and hence

$$\sqrt{N} (\hat{\tau}_{t,s}^{\text{CF}} - \tau) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^{\tau,*}), \quad V_{t,s}^{\tau,*} := \text{Var}(\varphi_{t,s}(D; \vartheta^*) - \varphi_{t,s}^0(D; \vartheta^{0,*})).$$

Lemma A.8 (Cross-fitted ℓ_1 -penalized nuisance rates (adapted from Hou et al. [2025], Lemma A21)). *Fix a constant number of folds K_f . Let the full sample $\mathcal{D} = \{D_1, \dots, D_N\}$ be partitioned into folds $\mathcal{D} = \bigcup_{k=1}^{K_f} \mathcal{D}_k$, and let $\mathcal{D}^{(-k)} = \mathcal{D} \setminus \mathcal{D}_k$ denote the out-of-fold data. For each fold k , let $\hat{\gamma}^{(-k)}$ be an estimator trained on $\mathcal{D}^{(-k)}$ (hence independent of $\mathcal{D}_k$). For an observation index $i \in \{1, \dots, N\}$, let $\text{fold}(i) \in \{1, \dots, K_f\}$ denote the fold index such that $D_i \in \mathcal{D}_{\text{fold}(i)}$; we use $\hat{\gamma}^{(-\text{fold}(i))}$ to denote the out-of-fold nuisance estimate used when evaluating the loss contribution from D_i . Consider the cross-fitted ℓ_1 -penalized estimator*

$$\hat{\beta} \in \arg \min_{\beta \in \mathbb{R}^p} \left\{ \frac{1}{K_f} \sum_{k=1}^{K_f} \ell_k(\beta; \hat{\gamma}^{(-k)}) + \lambda \|\beta\|_1 \right\}, \quad \ell_k(\beta; \gamma) := \tilde{E}_k[\ell(D; \beta, \gamma)],$$

where $\ell(D; \beta, \gamma)$ is convex in β . Define also the fold-averaged empirical risk

$$L_N(\beta) := \frac{1}{K_f} \sum_{k=1}^{K_f} \ell_k(\beta; \hat{\gamma}^{(-k)}).$$

Let $\bar{\beta}$ be the population minimizer at a population nuisance target $\bar{\gamma}$,

$$\bar{\beta} \in \arg \min_{\beta \in \mathbb{R}^p} \mathbb{E}[\ell(D; \beta, \bar{\gamma})], \quad s_\beta := \|\bar{\beta}\|_0.$$

Assume the following high-dimensional regularity conditions (cf. Assumption 6 of Hou et al. [2025] and the RSC framework of Negahban et al. [2012]):

- (a) (Sub-Gaussian covariates) Each coordinate of $\phi(\mathbf{X})$ is sub-Gaussian with $\max_{1 \leq j \leq p} \|\phi_j(\mathbf{X})\|_{\psi_2} \leq K_\phi$ (allowing $\|\phi(\mathbf{X})\|_\infty$ to grow at the usual $\sqrt{\log p}$ rate).
- (b) (Response/link tails and smoothness) Y is uniformly sub-Gaussian (or bounded), and the GLM mean function and its derivative are uniformly bounded on a neighborhood of $\bar{\beta}$.
- (c) (Identifiability) The population Gram matrix is non-degenerate: $\lambda_{\min}(\mathbb{E}[\phi(\mathbf{X})\phi(\mathbf{X})^T]) \geq c > 0$.
- (d) (Weight regularity) Any weights induced by γ (e.g., density-ratio type weights) have finite moments and are locally Lipschitz in γ on a neighborhood of $\bar{\gamma}$ (sufficient for the score stability bound used below).
- (e) (Restricted strong convexity) With probability tending to 1, $L_N(\cdot)$ satisfies an RSC inequality on the standard cone associated with $\text{supp}(\bar{\beta})$.

In our FACE-HD setting, we impose these regularity conditions in addition to Assumptions A.1, A.2, and A.3 (noting that (a) matches Assumption A.1(a), while (d) is ensured by the overlap/weight-moment assumptions and local smoothness). Choose $\lambda \asymp \sqrt{\log(p)/N}$. Then:

Remark A.3 (On the choice of N in λ). The scaling $\lambda \asymp \sqrt{\log(p)/N}$ is stated in terms of the total sample size N for notational simplicity. In applications where the nuisance loss for a given arm/site pair is effectively averaged over an arm-specific subsample (e.g., treated units in the source), one may equivalently write $\lambda \asymp \sqrt{\log(p)/N_{\text{eff}}}$ where N_{eff} is the corresponding effective sample size. Under the sampling-fraction regime and overlap/positivity conditions stated above, $N_{\text{eff}} \asymp N$, so the rate statements are unchanged.

1. (General)

$$\|\hat{\beta} - \bar{\beta}\|_2 = O_p\left(\sqrt{\frac{s_\beta \log(p)}{N}}\right) + O_p\left(\sup_{1 \leq k \leq K_f} \|\hat{\gamma}^{(-k)} - \bar{\gamma}\|_2\right).$$

2. (Special) If, for each fold k , the conditional foldwise score at $\bar{\beta}$ satisfies

$$\mathbb{E}\left[\nabla_{\beta} \ell(D; \bar{\beta}, \hat{\gamma}^{(-k)}) \mid \mathcal{D}^{(-k)}\right] = \mathbf{0},$$

then the plug-in score term disappears and

$$\|\hat{\beta} - \bar{\beta}\|_2 = O_p\left(\sqrt{\frac{s_\beta \log(p)}{N}}\right).$$

Moreover, we can convert the ℓ_2 -rate to an ℓ_1 -rate by an explicit cone argument. Let $\Delta := \hat{\beta} - \bar{\beta}$ and $\mathcal{J} := \text{supp}(\bar{\beta})$ with $|\mathcal{J}| = s_\beta$. From the proof below (see the KKT-based cone inequality leading to (2)), on a high-probability event we have

$$\|\Delta_{\mathcal{J}^c}\|_1 \leq 3\|\Delta_{\mathcal{J}}\|_1 + \frac{2}{\lambda}\|B\|_\infty\|\Delta\|_1,$$

where $\|B\|_\infty$ is the plug-in score term induced by cross-fitting. With $\lambda \asymp \sqrt{\log(p)/N}$ and $\|B\|_\infty = O_p(\sqrt{\log(p)/N})$ (proved below), we have $(2/\lambda)\|B\|_\infty \leq 1/2$ w.p.a.1, and hence

$$\|\Delta\|_1 = \|\Delta_{\mathcal{J}}\|_1 + \|\Delta_{\mathcal{J}^c}\|_1 \leq 4\|\Delta_{\mathcal{J}}\|_1 \leq 4\sqrt{s_\beta}\|\Delta\|_2.$$

Combining this with the ℓ_2 bounds above yields the corresponding ℓ_1 -rate:

$$\|\hat{\beta} - \bar{\beta}\|_1 = O_p\left(s_\beta \sqrt{\frac{\log(p)}{N}}\right) + O_p\left(\sqrt{s_\beta} \sup_{1 \leq k \leq K_f} \|\hat{\gamma}^{(-k)} - \bar{\gamma}\|_2\right),$$

with the second term dropping in the special case.

Instantiation for FACE-HD calibrated losses. In our application, β is instantiated as either α^1 or γ_1 . If Assumption A.3(d) holds (as in the foldwise score-centering condition used by Hou et al. [2025]), then we may invoke the (**Special**) case above and obtain

$$\|\hat{\alpha}_{t,s}^{1,(-k)} - \alpha^{1,*}\|_q = O_p\left(s_\alpha^{1-1/q} \sqrt{\frac{\log(p)}{N}}\right), \quad \|\hat{\gamma}_{s,1}^{(-k)} - \gamma_1^*\|_q = O_p\left(s_\gamma^{1-1/q} \sqrt{\frac{\log(p)}{N}}\right), \quad q \in \{1, 2\}.$$

Proof. We give a detailed argument in our notation; it follows the structure of Hou et al. [2025] (proof of Lemma A21). Write the (random) fold-averaged empirical loss as

$$L_N(\beta) := \frac{1}{K_f} \sum_{k=1}^{K_f} \ell_k(\beta; \hat{\gamma}^{(-k)}), \quad \hat{\beta} \in \arg \min_{\beta} \{L_N(\beta) + \lambda \|\beta\|_1\}.$$

Let $\Delta := \hat{\beta} - \bar{\beta}$, and let $\mathcal{J} := \text{supp}(\bar{\beta})$ with $|\mathcal{J}| = s_\beta$.

Step 1 (Basic inequality). By optimality,

$$L_N(\bar{\beta} + \Delta) + \lambda \|\bar{\beta} + \Delta\|_1 \leq L_N(\bar{\beta}) + \lambda \|\bar{\beta}\|_1,$$

so

$$L_N(\bar{\beta} + \Delta) - L_N(\bar{\beta}) \leq \lambda(\|\bar{\beta}\|_1 - \|\bar{\beta} + \Delta\|_1) \leq \lambda(\|\Delta_{\mathcal{J}}\|_1 - \|\Delta_{\mathcal{J}^c}\|_1).$$

Step 2 (RSC expansion). By convexity,

$$L_N(\bar{\beta} + \Delta) - L_N(\bar{\beta}) \geq \nabla L_N(\bar{\beta})^T \Delta + \mathcal{R}_N(\Delta),$$

where the remainder $\mathcal{R}_N(\Delta)$ satisfies the RSC lower bound (Assumption A.3(a) as applied to this loss)

$$\mathcal{R}_N(\Delta) \geq \kappa_1 \|\Delta\|_2^2 - \kappa_2 \frac{\log(p)}{N} \|\Delta\|_1^2$$

on the standard restricted cone. Combining with Step 1 yields

$$\kappa_1 \|\Delta\|_2^2 \leq \|\nabla L_N(\bar{\beta})\|_\infty \|\Delta\|_1 + \lambda(\|\Delta_{\mathcal{J}}\|_1 - \|\Delta_{\mathcal{J}^c}\|_1) + \kappa_2 \frac{\log(p)}{N} \|\Delta\|_1^2. \quad (1)$$

Step 3 (Score decomposition and choice of λ). Decompose the score

$$\nabla L_N(\bar{\beta}) = \underbrace{(\nabla L_N(\bar{\beta}) - \mathbb{E}[\nabla L_N(\bar{\beta}) \mid \{\mathcal{D}^{(-k)}\}_{k=1}^{K_f}])}_{\text{stochastic term } A} + \underbrace{\mathbb{E}[\nabla L_N(\bar{\beta}) \mid \{\mathcal{D}^{(-k)}\}_{k=1}^{K_f}]}_{\text{plug-in term } B}.$$

By cross-fitting, conditional on $\{\mathcal{D}^{(-k)}\}$, each fold average $\tilde{E}_k[\nabla_{\beta} \ell(D; \bar{\beta}, \hat{\gamma}^{(-k)})]$ is an average of independent sub-exponential random vectors with uniformly controlled coordinates, so a union bound with Bernstein/Hoeffding inequalities gives

$$\|A\|_\infty = O_p\left(\sqrt{\frac{\log(p)}{N}}\right).$$

Choose $\lambda \asymp \sqrt{\log(p)/N}$ so that $\lambda \geq 2\|A\|_\infty$ w.p.a.1.

For the plug-in term, note that

$$B - \mathbb{E}[\nabla_{\beta} \ell(D; \bar{\beta}, \bar{\gamma})] = \mathbb{E}[\nabla_{\beta} \ell(D; \bar{\beta}, \hat{\gamma}^{(-\text{fold}(D))}) - \nabla_{\beta} \ell(D; \bar{\beta}, \bar{\gamma}) \mid \{\mathcal{D}^{(-k)}\}],$$

and the weight-regularity/Lipschitz condition implies

$$\|B - \mathbb{E}[\nabla_{\beta} \ell(D; \bar{\beta}, \bar{\gamma})]\|_2 \leq C \sup_{1 \leq k \leq K_f} \|\hat{\gamma}^{(-k)} - \bar{\gamma}\|_2.$$

In the special case, $\mathbb{E}[\nabla_{\beta} \ell(D; \bar{\beta}, \hat{\gamma}^{(-k)}) \mid \mathcal{D}^{(-k)}] = 0$ for all k , hence $B = \mathbf{0}$.

Step 4 (Cone property and ℓ_2 rate). Plugging $\|\nabla L_N(\bar{\beta})\|_\infty \leq \|A\|_\infty + \|B\|_\infty$ into (1) and using $\lambda \geq 2\|A\|_\infty$ yields the cone inequality

$$\|\Delta_{\mathcal{J}^c}\|_1 \leq 3\|\Delta_{\mathcal{J}}\|_1 + \frac{2}{\lambda} \|B\|_\infty \|\Delta\|_1. \quad (2)$$

On the event $\{(2/\lambda)\|B\|_\infty \leq 1/2\}$, (2) implies $\|\Delta_{\mathcal{J}^c}\|_1 \leq 7\|\Delta_{\mathcal{J}}\|_1$ and hence $\|\Delta\|_1 \leq 8\|\Delta_{\mathcal{J}}\|_1 \leq 8\sqrt{s_\beta}\|\Delta\|_2$. Plugging this back into the RSC lower bound (Assumption A.3(a) as applied to this loss) absorbs the $\kappa_2(\log(p)/N)\|\Delta\|_1^2$ term and yields

$$\|\Delta\|_2 = O_p(\sqrt{s_\beta}\lambda) + O_p\left(\sup_{1 \leq k \leq K_f} \|\hat{\gamma}^{(-k)} - \bar{\gamma}\|_2\right).$$

This is the general-case ℓ_2 rate. In the special case, $B = \mathbf{0}$ and the second term drops.

Step 5 (ℓ_1 rate). From the cone property and $\|\Delta_{\mathcal{J}}\|_1 \leq \sqrt{s_\beta}\|\Delta\|_2$,

$$\|\Delta\|_1 = O_p(s_\beta\lambda) + O_p\left(\sqrt{s_\beta} \sup_{1 \leq k \leq K_f} \|\hat{\gamma}^{(-k)} - \bar{\gamma}\|_2\right),$$

with the second term dropping in the special case.

Finally, for FACE-HD, the **(Special)** conclusion may be invoked for $\beta = \alpha^1$ and $\beta = \gamma_1$ only when the foldwise score-centering condition is assumed/verified as in Assumption A.3(d). Otherwise, one should use the **(General)** bounds, which explicitly propagate the plug-in nuisance error. $\square$

Lemma A.9 (Orthogonality from Calibrated Losses). *Under Assumption A.3(c), let $\vartheta^* = (\alpha^{1,*}, \gamma_1^*)$ be the population targets of the calibrated losses in the two-level algorithm (main manuscript appendix, Section B; Equations (50) and (51)):*

$$\begin{aligned} L_\gamma(\gamma; \alpha^{1,*}) &= \mathbb{E}_t [\nabla_\alpha \psi(\phi(\mathbf{X}); \alpha^{1,*})^T \gamma] + \mathbb{E}_s [I(A=1)e^{-\phi(\mathbf{X})^T \gamma} \psi'(\mathcal{T}(\phi(\mathbf{X})^T \alpha^{1,*}))] \\ L_\alpha(\alpha; \gamma_1^*) &= \mathbb{E}_s [w_\tau(\mathbf{X}; \gamma_1^*) (-Y \cdot (\phi(\mathbf{X})^T \alpha) + \Psi(\phi(\mathbf{X})^T \alpha))] \end{aligned}$$

(Ignoring regularization terms, which vanish in population). For canonical-link GLMs,

$$\nabla_\alpha \psi(\phi(\mathbf{X}); \alpha) = \phi(\mathbf{X}) \psi'(\phi(\mathbf{X})^T \alpha).$$

Remark A.4 (Penalization and approximate orthogonality). *The population moment conditions below are derived from the unpenalized population losses. In finite samples, the ℓ_1 penalty induces a KKT subgradient term of order λ in the estimating equations. Under the nuisance rates in Lemma A.8 and the sparsity scaling used in Theorem A.3, this penalty-induced perturbation contributes at most an $o_p(N^{-1/2})$ term to the final DR remainder and therefore does not affect the first-order limit theory.*

Under asymptotically inactive truncation (Assumption A.2(c)), the population first-order optimality conditions are therefore equivalent to the following moment conditions at ϑ^* :

$$\begin{aligned} (i) \quad & \mathbb{E}_t [\nabla_\alpha \psi(\phi(\mathbf{X}); \alpha^{1,*})] - \mathbb{E}_s [w(\mathbf{X}; \gamma_1^*) \nabla_\alpha \psi(\phi(\mathbf{X}); \alpha^{1,*})] = \mathbf{0} \\ (ii) \quad & \mathbb{E}_s [w(\mathbf{X}; \gamma_1^*) \phi(\mathbf{X}) \{Y - \psi(\phi(\mathbf{X})^T \alpha^{1,*})\}] = \mathbf{0}. \end{aligned}$$

Proof. We derive the moment conditions by taking the gradients of the population loss functions L_γ and L_α with respect to the parameter being optimized and setting them to zero at the minimum ϑ^* .

Condition (i) from FOC of $L_\gamma(\gamma; \alpha^{1,*})$ w.r.t. γ : The gradient of $L_\gamma(\gamma; \alpha^{1,*})$ with respect to γ is:

$$\begin{aligned} \nabla_\gamma L_\gamma(\gamma; \alpha^{1,*}) &= \mathbb{E}_t [\nabla_\alpha \psi(\phi(\mathbf{X}); \alpha^{1,*})] \\ &\quad + \mathbb{E}_s [I(A=1)(-\phi(\mathbf{X}))e^{-\phi(\mathbf{X})^T \gamma} \psi'(\mathcal{T}(\phi(\mathbf{X})^T \alpha^{1,*}))] \end{aligned}$$

Setting this gradient to zero at $\gamma = \gamma_1^*$ gives:

$$\mathbb{E}_t [\nabla_\alpha \psi(\phi(\mathbf{X}); \alpha^{1,*})] = \mathbb{E}_s [\phi(\mathbf{X}) w(\mathbf{X}; \gamma_1^*) \psi'(\mathcal{T}(\phi(\mathbf{X})^T \alpha^{1,*}))].$$

This equation is constructed to enforce the moment condition (i). Note that for standard GLMs with canonical links,

$$\nabla_\alpha \psi(\phi(\mathbf{X}); \alpha) = \phi(\mathbf{X}) \psi'(\phi(\mathbf{X})^T \alpha).$$

Truncation Error Analysis: Define the event $\mathcal{E}_{\alpha,N} = \{|\phi(\mathbf{X})^T \alpha^{1,*}| \leq M_{\tau,N}\}$. By Assumption A.2(c), $P(\mathcal{E}_{\alpha,N}^c | R=s) = o(N^{-1})$. On $\mathcal{E}_{\alpha,N}$, $\mathcal{T}(\phi(\mathbf{X})^T \alpha^{1,*}) = \phi(\mathbf{X})^T \alpha^{1,*}$, so $\psi'(\mathcal{T}(\cdot)) = \psi'(\cdot)$ exactly. Therefore, replacing $\psi'(\mathcal{T}(\phi(\mathbf{X})^T \alpha^{1,*}))$ by $\psi'(\phi(\mathbf{X})^T \alpha^{1,*})$ in the population FOC changes the source-side moment by at most an $o(1)$ term, and condition (i) holds asymptotically.

Condition (ii) from FOC of $L_\alpha(\alpha; \gamma_1^*)$ w.r.t. α : The loss L_α is the population weighted GLM log-likelihood. Its gradient with respect to α is:

$$\nabla_\alpha L_\alpha(\alpha; \gamma_1^*) = \mathbb{E}_s [w_\tau(\mathbf{X}; \gamma_1^*) (-Y \cdot \phi(\mathbf{X}) + \nabla_\alpha \Psi(\phi(\mathbf{X})^T \alpha))]$$

Using the GLM identity $\nabla_\alpha \Psi(u) = \psi(u)\phi(\mathbf{X})$ where $u = \phi(\mathbf{X})^T \alpha$:

$$\nabla_\alpha L_\alpha(\alpha; \gamma_1^*) = \mathbb{E}_s [w_\tau(\mathbf{X}; \gamma_1^*) \phi(\mathbf{X}) (\psi(\phi(\mathbf{X})^T \alpha) - Y)]$$

Setting this gradient to zero at $\alpha = \alpha^{1,*}$ gives:

$$\mathbb{E}_s [w_\tau(\mathbf{X}; \gamma_1^*) \phi(\mathbf{X}) (Y - \psi(\phi(\mathbf{X})^T \alpha^{1,*}))] = 0.$$

Finally, under Assumption A.2(c), truncation is asymptotically inactive at γ_1^* , so $w_\tau(\mathbf{X}; \gamma_1^*) = w(\mathbf{X}; \gamma_1^*)$ w.p.a.1. This yields condition (ii). $\square$

Lemma A.10 (Bias-Variance Bounds). *Let $\vartheta_n = (\alpha_n^1, \gamma_{1,n})$ be a sequence of nuisance parameters converging to $\vartheta^* = (\alpha^{1,*}, \gamma_1^*)$.*

1. (**Bias Bound - Double Robustness**) *If ϑ^* minimizes the population calibrated losses (Lemma A.9 holds), then:*

$$\begin{aligned} |\mathbb{E}[\tilde{\varphi}_{t,s}(D; \vartheta_n)] - \mathbb{E}[\tilde{\varphi}_{t,s}(D; \vartheta^*)]| &\leq C \cdot (\|\alpha_n^1 - \alpha^{1,*}\|_2^2 + \|\gamma_{1,n} - \gamma_1^*\|_2^2 \\ &\quad + \|\alpha_n^1 - \alpha^{1,*}\|_2 \cdot \|\gamma_{1,n} - \gamma_1^*\|_2) \end{aligned}$$

2. (**Variance Bound**)

$$\text{Var}(\tilde{\varphi}_{t,s}(D; \vartheta_n) - \tilde{\varphi}_{t,s}(D; \vartheta^*)) \leq C \cdot (\|\alpha_n^1 - \alpha^{1,*}\|_2^2 + \|\gamma_{1,n} - \gamma_1^*\|_2^2)$$

Proof. We analyze the bias term $\mathbb{E}[\tilde{\varphi}_{t,s}(D; \vartheta_n) - \tilde{\varphi}_{t,s}(D; \vartheta^*)]$. Let $\Delta_\alpha = \alpha_n^1 - \alpha^{1,*}$ and $\Delta_\gamma = \gamma_{1,n} - \gamma_1^*$. We decompose the bias using an intermediate step:

$$\text{Bias} = \underbrace{\mathbb{E}[\tilde{\varphi}_{t,s}(D; \alpha_n^1, \gamma_{1,n}) - \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_{1,n})]}_{\text{Bias from } \alpha} + \underbrace{\mathbb{E}[\tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_{1,n}) - \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)]}_{\text{Bias from } \gamma}$$

Step 1 (Bias from α : mean-value expansion + orthogonality). Define the path $h_\alpha(t) := \mathbb{E}[\tilde{\varphi}_{t,s}(D; \alpha^{1,*} + t\Delta_\alpha, \gamma_{1,n})]$ for $t \in [0, 1]$. By the fundamental theorem of calculus (and dominated convergence, justified by Assumption A.2(b) and local boundedness of the derivatives), we have

$$h_\alpha(1) - h_\alpha(0) = \int_0^1 h'_\alpha(t) dt = \int_0^1 \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*} + t\Delta_\alpha, \gamma_{1,n})]^T \Delta_\alpha dt.$$

Hence

$$\text{Bias from } \alpha = (\mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)])^T \Delta_\alpha + \int_0^1 \{\mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*} + t\Delta_\alpha, \gamma_{1,n})] - \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)]\}^T \Delta_\alpha dt.$$

The first term is zero by Lemma A.9(i). For the second term, we use Cauchy-Schwarz:

$$|\text{Bias from } \alpha| \leq \|\Delta_\alpha\|_2 \int_0^1 \|\mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*} + t\Delta_\alpha, \gamma_{1,n})] - \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)]\|_2 dt.$$

The mapping $\vartheta \mapsto \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \vartheta)]$ is locally Lipschitz in ϑ by the explicit derivative formula below, the bounded-slope/Lipschitz condition on ψ' (Assumption A.2(b)(i)), and the bounded-weights/moment conditions (Assumption A.2(a,b)). Therefore, for ϑ_n sufficiently close to ϑ^* ,

$$\|\mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*} + t\Delta_\alpha, \gamma_{1,n})] - \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)]\|_2 \leq C (\|\Delta_\alpha\|_2 + \|\Delta_\gamma\|_2),$$

uniformly over $t \in [0, 1]$, and hence

$$|\text{Bias from } \alpha| \leq C (\|\Delta_\alpha\|_2^2 + \|\Delta_\alpha\|_2 \|\Delta_\gamma\|_2).$$

Step 2 (Bias from γ : mean-value expansion + orthogonality). Similarly define $h_\gamma(t) := \mathbb{E}[\tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^* + t\Delta_\gamma)]$. Then

$$\text{Bias from } \gamma = \int_0^1 \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^* + t\Delta_\gamma)]^T \Delta_\gamma dt.$$

Adding and subtracting $\mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)]$, the leading term is zero by Lemma A.9(ii), and the remainder is bounded (again by local Lipschitz continuity and Cauchy-Schwarz) as

$$|\text{Bias from } \gamma| \leq C \|\Delta_\gamma\|_2^2.$$

Step 3 (Collecting). Combining the two displays yields the stated second-order bias bound.

Derivative identities used above. For example, the gradient w.r.t. α at $(\alpha^{1,*}, \gamma_1^*)$ is:

$$\mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)] = \mathbb{E}_t[\nabla_\alpha \psi(\cdot; \alpha^{1,*})] - \mathbb{E}_s[w(\mathbf{X}; \gamma_1^*) \nabla_\alpha \psi(\cdot; \alpha^{1,*})]$$

where $w(\mathbf{X}; \gamma) = I(A=1)/e^{\phi(\mathbf{X})^T \gamma}$. This is exactly $\mathbf{0}$ by Lemma A.9(i). Similarly, the gradient w.r.t. γ is:

$$\begin{aligned} \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)] &= \mathbb{E}_s[\nabla_\gamma w(\mathbf{X}; \gamma_1^*) \{Y - \psi(\phi(\mathbf{X}); \alpha^{1,*})\}] \\ &= -\mathbb{E}_s[w(\mathbf{X}; \gamma_1^*) \phi(\mathbf{X}) \{Y - \psi(\phi(\mathbf{X}); \alpha^{1,*})\}] \end{aligned}$$

This is exactly $\mathbf{0}$ by Lemma A.9(ii). This proves part 1.

Part 2 (Variance): explicit L_2 -Lipschitz bound. Write the difference as

$$\tilde{\varphi}_{t,s}(D; \vartheta_n) - \tilde{\varphi}_{t,s}(D; \vartheta^*) = T_t(D) + T_s(D),$$

where

$$T_t(D) := \frac{I(R=t)}{\rho_t} \{ \psi(\phi(\mathbf{X}); \alpha_n^1) - \psi(\phi(\mathbf{X}); \alpha^{1,*}) \},$$

and

$$\begin{aligned} T_s(D) &:= \frac{I(R=s)}{\rho_s} \left(w(\mathbf{X}; \gamma_{1,n}) \{Y - \psi(\phi(\mathbf{X}); \alpha_n^1)\} - w(\mathbf{X}; \gamma_1^*) \{Y - \psi(\phi(\mathbf{X}); \alpha^{1,*})\} \right) \\ &= \frac{I(R=s)}{\rho_s} \left(\underbrace{\{w(\mathbf{X}; \gamma_{1,n}) - w(\mathbf{X}; \gamma_1^*)\} \{Y - \psi(\phi(\mathbf{X}); \alpha^{1,*})\}}_{T_{s,1}(D)} + \underbrace{w(\mathbf{X}; \gamma_{1,n}) \{\psi(\phi(\mathbf{X}); \alpha^{1,*}) - \psi(\phi(\mathbf{X}); \alpha_n^1)\}}_{T_{s,2}(D)} \right). \end{aligned}$$

By $(a+b+c)^2 \leq 3(a^2+b^2+c^2)$ and $\text{Var}(Z) \leq \mathbb{E}[Z^2]$,

$$\text{Var}(\tilde{\varphi}_{t,s}(D; \vartheta_n) - \tilde{\varphi}_{t,s}(D; \vartheta^*)) \leq 3\mathbb{E}[T_t(D)^2] + 3\mathbb{E}[T_{s,1}(D)^2] + 3\mathbb{E}[T_{s,2}(D)^2].$$

For T_t and $T_{s,2}$, the bounded-slope assumption on ψ (Assumption A.2(b)(i)) implies $|\psi(u_1) - \psi(u_2)| \leq L_\psi |u_1 - u_2|$, hence

$$\mathbb{E}[T_t(D)^2] + \mathbb{E}[T_{s,2}(D)^2] \leq C \mathbb{E}[\|\phi(\mathbf{X})\|_2^2] \|\Delta_\alpha\|_2^2,$$

using the weight-moment condition (Assumption A.2(a)(ii)) and finite second moments of $\phi(\mathbf{X})$ (Assumption A.1(a)). For $T_{s,1}$, local L_2 -Lipschitz continuity of $w(\mathbf{X}; \gamma)$ in γ (implied by the exponential form of w , the overlap/weight-moment assumptions, and the moment condition in Assumption A.2(b)) yields

$$\mathbb{E}[T_{s,1}(D)^2] \leq C \|\Delta_\gamma\|_2^2.$$

Combining the bounds gives the stated variance inequality. $\square$

Lemma A.11 (Double Robustness of Bias Terms). *Let $\Delta_\alpha = \alpha_n^1 - \alpha^{1,*}$ and $\Delta_\gamma = \gamma_{1,n} - \gamma_1^*$. Define the two first-order bias components:*

$$\begin{aligned} \Psi_1(\alpha_n^1, \gamma_1^*) &:= \mathbb{E}[\tilde{\varphi}_{t,s}(D; \alpha_n^1, \gamma_1^*) - \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)] \\ \Psi_2(\gamma_{1,n}, \alpha^{1,*}) &:= \mathbb{E}[\tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_{1,n}) - \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)] \end{aligned}$$

Under the conditions:

1. If Lemma A.9(i) holds (from calibration w.r.t. γ), then $\Psi_1(\alpha_n^1, \gamma_1^*) = O(\|\Delta_\alpha\|_2^2)$.

2. If Lemma A.9(ii) holds (from calibration w.r.t. α), then $\Psi_2(\gamma_{1,n}, \alpha^{1,*}) = O(\|\Delta_\gamma\|_2^2)$.

Note: For identification of the target parameter, Theorem A.3 still requires at least one nuisance model to be correctly specified so that $\mu^1(\vartheta^*) = \mu^1$. This is formalized in Lemma A.1.

Proof. 1. Analyzing Ψ_1 using Lemma A.9(i) (mean-value expansion). Let $h_1(t) := \mathbb{E}[\tilde{\varphi}_{t,s}(D; \alpha^{1,*} + t\Delta_\alpha, \gamma_1^*)]$ for $t \in [0, 1]$. Then $\Psi_1 = h_1(1) - h_1(0)$ and

$$\Psi_1 = \int_0^1 h_1'(t) dt = \int_0^1 \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*} + t\Delta_\alpha, \gamma_1^*)]^T \Delta_\alpha dt.$$

Add and subtract $\mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)]$ inside the integral to get

$$\Psi_1 = (\mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)])^T \Delta_\alpha + \int_0^1 \{\mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*} + t\Delta_\alpha, \gamma_1^*)] - \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)]\}^T \Delta_\alpha dt.$$

The first term is zero by Lemma A.9(i). For the second term, we bound the gradient difference. From the definition of $\tilde{\varphi}_{t,s}$ and the canonical-link GLM identity $\nabla_\alpha \psi(\phi(\mathbf{X}); \alpha) = \phi(\mathbf{X})\psi'(\phi(\mathbf{X})^T \alpha)$,

$$\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha, \gamma_1^*) = \frac{I(R=t)}{\rho_t} \phi(\mathbf{X})\psi'(\phi(\mathbf{X})^T \alpha) - \frac{I(R=s)}{\rho_s} w(\mathbf{X}; \gamma_1^*) \phi(\mathbf{X})\psi'(\phi(\mathbf{X})^T \alpha).$$

Using the Lipschitz property of ψ' (Assumption A.2(b)(i)) and the weight-moment condition (Assumption A.2(a)(ii)), we have

$$\|\mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*} + t\Delta_\alpha, \gamma_1^*)] - \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)]\|_2 \leq C \|\Delta_\alpha\|_2,$$

uniformly over $t \in [0, 1]$, where $C < \infty$ depends on ρ_t, ρ_s , the moment bound on $\phi(\mathbf{X})$ (Assumption A.1(a)), and $L_{\psi'}$. Therefore, by Cauchy–Schwarz,

$$|\Psi_1| \leq \int_0^1 C \|\Delta_\alpha\|_2 \cdot \|\Delta_\alpha\|_2 dt = O(\|\Delta_\alpha\|_2^2).$$

2. Analyzing Ψ_2 using Lemma A.9(ii) (mean-value expansion). Let $h_2(t) := \mathbb{E}[\tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^* + t\Delta_\gamma)]$. Then $\Psi_2 = h_2(1) - h_2(0)$ and

$$\Psi_2 = \int_0^1 \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^* + t\Delta_\gamma)]^T \Delta_\gamma dt.$$

Add and subtract the gradient at γ_1^* :

$$\Psi_2 = (\mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)])^T \Delta_\gamma + \int_0^1 \{\mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^* + t\Delta_\gamma)] - \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)]\}^T \Delta_\gamma dt.$$

The first term is zero by Lemma A.9(ii). For the remainder, recall $w(\mathbf{X}; \gamma) = I(A=1) \exp(-\phi(\mathbf{X})^T \gamma)$ so that $\nabla_\gamma w(\mathbf{X}; \gamma) = -w(\mathbf{X}; \gamma) \phi(\mathbf{X})$. Thus

$$\nabla_\gamma \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma) = \frac{I(R=s)}{\rho_s} \nabla_\gamma w(\mathbf{X}; \gamma) \{Y - \psi(\phi(\mathbf{X}); \alpha^{1,*})\}.$$

Under Assumption A.2(a,b), the mapping $\gamma \mapsto \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma)]$ is locally Lipschitz in γ , hence

$$\|\mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^* + t\Delta_\gamma)] - \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)]\|_2 \leq C \|\Delta_\gamma\|_2$$

uniformly over $t \in [0, 1]$. Therefore $|\Psi_2| \leq \int_0^1 C \|\Delta_\gamma\|_2^2 dt = O(\|\Delta_\gamma\|_2^2)$.

Clarification on Double Robustness: The key insight is that the calibrated loss functions are designed such that:

- Lemma A.9(i) holds *by construction* of L_γ at any ϑ^* that minimizes L_γ , regardless of whether the weighting model is correct.
- Lemma A.9(ii) holds *by construction* of L_α at any ϑ^* that minimizes L_α , regardless of whether the outcome model is correct.

Thus the first-order sensitivity to each nuisance can be removed by calibration; combined with model correctness of at least one nuisance for identification, this yields the double-robust conclusion in Theorem A.3. $\square$

B Aggregated Estimator: FACE-HD

B.1 Setup and Notation for Aggregation

Following Section 4.3 in the main manuscript, we aggregate the source-assisted estimators $\{\hat{\mu}_{t,s_j}^1\}_{j=1}^K$ with the target-only estimator $\hat{\mu}_{ot}^1$ using an adaptive weighting scheme. Let K be the number of source sites. Define the global sample size across all participating sites as

$$N_{\text{all}} := N_t + \sum_{j=1}^K N_{s_j},$$

and assume $N_t/N_{\text{all}} \rightarrow \rho_t^{\text{all}} \in (0, 1)$ and $N_{s_j}/N_{\text{all}} \rightarrow \rho_{s_j}^{\text{all}} \in (0, 1)$ for each fixed j .

Notation Correspondence with Main Manuscript:

- $\hat{\mu}_{ot}^1$: Target-only estimator for $\mu^1 = \mathbb{E}_t[Y(1)]$ (uses only target site data)
- $\hat{\mu}_{t,s_j}^1$: Source-assisted estimator using source site s_j (from Section 2.3)
- $\hat{\mu}_{\text{agg}}^1$: Aggregated FACE-HD estimator
- η_j : Weight for source site s_j
- $\varphi_{ot,i}$: Population IF contribution for the target-only estimator (population analogue of $\hat{\varphi}_{ot,i}$ in the main manuscript)

The FACE-HD aggregated estimator (Equation (22) in main manuscript) is:

$$\hat{\mu}_{\text{agg}}^1 = \left(1 - \sum_{j=1}^K \eta_j\right) \hat{\mu}_{ot}^1 + \sum_{j=1}^K \eta_j \hat{\mu}_{t,s_j}^1$$

which can be rewritten as:

$$\hat{\mu}_{\text{agg}}^1 = \hat{\mu}_{ot}^1 + \sum_{j=1}^K \eta_j \left(\hat{\mu}_{t,s_j}^1 - \hat{\mu}_{ot}^1 \right)$$

where $\hat{\eta} = (\hat{\eta}_1, \dots, \hat{\eta}_K)^T$ are adaptive weights minimizing the estimated asymptotic variance with a selection penalty.

B.2 Structure of the Aggregated Asymptotic Covariance

Before turning to the penalized estimation of η , we record a structural result on the population covariance entering the aggregation objective. The result clarifies why cross-site covariance terms only couple through the *shared target sample* and justifies the block decomposition used implicitly in Equation (22) of the main manuscript.

For each source s_j , decompose the pairwise influence-function contribution into an outcome-regression part and a weighted-residual part:

$$\varphi_{t,s_j}(D; \vartheta_{s_j}^*) = \underbrace{\frac{I(R=t)}{\rho_t} \{ \psi(\phi(\mathbf{X}); \alpha_{t,s_j}^{1,*}) - \mu_t^1(\alpha_{t,s_j}^{1,*}) \}}_{=: \xi_{t,s_j}^{\text{OR}}(D)} + \underbrace{\frac{I(R=s_j)}{\rho_{s_j}} \{ w(\mathbf{X}; \gamma_{s_j,1}^*)(Y - \psi(\phi(\mathbf{X}); \alpha_{t,s_j}^{1,*})) - \delta_{s_j}^1(\vartheta_{s_j}^*) \}}_{=: \xi_{s_j}^{\text{IPW}}(D)}.$$

For the target-only estimator, denote analogously by $\varphi_{ot}(D)$ the AIPW influence function built from the target-site propensity π_{ot} and a target-only outcome model, and by $\xi_{ot}^{\text{OR}}(D)$ and $\xi_{ot}^{\text{IPW}}(D)$ its outcome and residual components.

Lemma B.1 (Block structure of the asymptotic covariance). *Suppose the pairwise estimators $\hat{\mu}_{ot}^1$ and $\{\hat{\mu}_{t,s_j}^1\}_{j=1}^K$ admit the influence-function representations given above, with $\mathbb{E}[\xi_{ot}^{\text{OR}}(D)] = 0$ and $\mathbb{E}[\xi_{ot}^{\text{IPW}}(D) \mid R = s_j] = 0$ (the latter by the calibration moment condition). Then, for any $j, k \in \{1, \dots, K\}$ with $j \neq k$:*

- (a) **(Source residuals are mutually uncorrelated.)** *Because observations with $R = s_j$ and $R = s_k$ are disjoint, $\mathbb{E}[\xi_{s_j}^{\text{IPW}}(D) \xi_{s_k}^{\text{IPW}}(D)] = 0$.*

- (b) **(Source residuals are uncorrelated with every OR component.)** By the indicator $I(R = s_j)$ and the target-site support of ξ_{t,s_k}^{OR} , $\mathbb{E}[\xi_{s_j}^{\text{IPW}}(D) \xi_{t,s_k}^{\text{OR}}(D)] = \mathbb{E}[\xi_{s_j}^{\text{IPW}}(D) \xi_{ot}^{\text{OR}}(D)] = 0$, for any j, k .
- (c) **(Cross-site correlation is driven solely by the shared target sample.)** Consequently,

$$\text{Cov}(\varphi_{t,s_j}(D), \varphi_{t,s_k}(D)) = \text{Cov}(\xi_{t,s_j}^{\text{OR}}(D), \xi_{t,s_k}^{\text{OR}}(D)), \quad j \neq k,$$

and similarly the covariance between φ_{ot} and any φ_{t,s_j} reduces to the covariance of their OR components evaluated on the target sample.

Proof. Parts (a) and (b) follow from $I(R = s_j)I(R = s_k) = 0$ for $j \neq k$, from $I(R = s_j)I(R = t) = 0$, and from $\mathbb{E}[\xi_{s_j}^{\text{IPW}}(D) | R = s_j] = 0$. Part (c) follows by expanding the covariance using the decomposition of each φ_{t,s_j} into its OR and IPW components and applying (a)–(b) to cancel the IPW cross-terms. $\square$

The block structure established by Lemma B.1 implies that the population Hessian H in the quadratic aggregation objective $L^*(\boldsymbol{\eta}) = \boldsymbol{\eta}^\top H \boldsymbol{\eta} - 2\mathbf{g}^\top \boldsymbol{\eta} + c$ decomposes as $H = H^{\text{OR}} + H^{\text{IPW}}$, where H^{IPW} is diagonal (containing only the per-source residual variances) and H^{OR} is the $K \times K$ Gram matrix induced by the shared target-sample OR components. This decomposition is the direct high-dimensional counterpart of Equation (11) in Han et al. [2025]: the first sum $\sum_{k_s \in \mathcal{S}} \eta_{k_s}^2 \hat{\sigma}_{k_s}^2 / n_{k_s}$ in the FACE aggregation objective corresponds to the diagonal H^{IPW} contribution, and the second sum $\sum_{k_t \in \mathcal{T}} \hat{h}_{k_t}(\boldsymbol{\eta})^\top \hat{\Sigma}_{k_t} / n_{k_t} \hat{h}_{k_t}(\boldsymbol{\eta})$ encodes the cross-source coupling through the shared target sample that corresponds to H^{OR} . The full penalized QP in Equation (22) of the main manuscript, like the FACE objective, does not admit a closed-form minimizer in general and is solved numerically. The following corollary records the explicit solution in the limiting regime where the target-sample coupling vanishes, which is the high-dimensional analogue of the same diagonal simplification implicit in the coordinate-wise structure of the FACE aggregation rule.

Corollary B.2 (Soft-thresholding closed form under diagonal Hessian). *Let $\Delta_j := \hat{\mu}_{t,s_j}^1 - \hat{\mu}_{ot}^1$ and $d_j := \hat{\mu}_{ot}^1 - \hat{\mu}_{t,s_j}^1 = -\Delta_j$. Suppose that, in addition to the hypotheses of Lemma B.1, the OR components are pairwise uncorrelated across source sites in the sense that*

$$\text{Cov}(\xi_{t,s_j}^{\text{OR}}(D), \xi_{t,s_k}^{\text{OR}}(D)) = 0 \quad \text{for all } j \neq k. \quad (3)$$

Then the population Hessian H is diagonal with entries

$$H_{jj} = \text{Var}(\Delta_j) = \sigma_j^2, \quad g_j = -\text{Cov}(\hat{\mu}_{ot}^1, \Delta_j),$$

where σ_j^2 collects the pairwise and target-only variance components present in Equation (22) of the main manuscript, and the penalized aggregation problem

$$\hat{\boldsymbol{\eta}} = \arg \min_{\boldsymbol{\eta} \in \mathbb{R}^K} \left\{ \boldsymbol{\eta}^\top \hat{H} \boldsymbol{\eta} - 2\hat{\mathbf{g}}^\top \boldsymbol{\eta} + \lambda \sum_{j=1}^K |\eta_j| d_j^2 \right\}$$

with $\hat{H}$ diagonal admits the coordinate-wise closed-form soft-thresholding solution

$$\hat{\eta}_j = \frac{1}{\hat{H}_{jj}} \text{sign}(\hat{g}_j) \max\{0, |\hat{g}_j| - \frac{\lambda}{2} d_j^2\} = \text{sign}(\hat{\eta}_j^{\text{OLS}}) \max\left\{0, |\hat{\eta}_j^{\text{OLS}}| - \frac{\lambda d_j^2}{2 \hat{H}_{jj}}\right\}, \quad (4)$$

where $\hat{\eta}_j^{\text{OLS}} := \hat{g}_j / \hat{H}_{jj}$ is the unpenalized coordinate minimizer. In particular, $\hat{\eta}_j = 0$ whenever $|\hat{\eta}_j^{\text{OLS}}| \leq \lambda d_j^2 / (2 \hat{H}_{jj})$, so the selection step of the aggregation becomes an explicit thresholding rule.

Proof. Under (3), Lemma B.1(c) gives $\text{Cov}(\Delta_j, \Delta_k) = \text{Cov}(\xi_{t,s_j}^{\text{OR}}, \xi_{t,s_k}^{\text{OR}}) + \text{Cov}(\xi_{s_j}^{\text{IPW}}, \xi_{s_k}^{\text{IPW}}) = 0$ for $j \neq k$, where the IPW cross-term vanishes by disjoint support. Hence the population (and plug-in empirical) Hessian is diagonal, $\hat{H} = \text{diag}(\hat{H}_{11}, \dots, \hat{H}_{KK})$.

Because $\hat{H}$ is diagonal and the penalty decouples across coordinates, the aggregation problem separates into K univariate problems:

$$\hat{\eta}_j = \arg \min_{\eta \in \mathbb{R}} \{ \hat{H}_{jj} \eta^2 - 2\hat{g}_j \eta + \lambda d_j^2 |\eta| \}. \quad (5)$$

Fix j and let $h := \hat{H}_{jj} > 0$, $g := \hat{g}_j$, $\ell := \lambda d_j^2/2 \geq 0$. The objective in (5) is strictly convex, so its minimizer is characterized by the subdifferential optimality condition

$$0 \in \partial(h\eta^2 - 2g\eta + 2\ell|\eta|) = \{2h\eta - 2g + 2\ell v : v \in \text{sgn}(\eta)\},$$

where $\text{sgn}(\eta) = \{1\}$ if $\eta > 0$, $\{-1\}$ if $\eta < 0$, and $[-1, 1]$ if $\eta = 0$.

Case $\eta > 0$. Optimality requires $h\eta - g + \ell = 0$, i.e., $\eta = (g - \ell)/h$. This is consistent with $\eta > 0$ iff $g > \ell$, in which case $\eta = (g - \ell)/h > 0$ minimizes the objective (by strict convexity).

Case $\eta < 0$. Optimality requires $h\eta - g - \ell = 0$, i.e., $\eta = (g + \ell)/h$. This is consistent with $\eta < 0$ iff $g < -\ell$, in which case $\eta = (g + \ell)/h < 0$.

Case $\eta = 0$. Optimality requires $0 \in 2h \cdot 0 - 2g + 2\ell \cdot [-1, 1]$, i.e., $g \in [-\ell, \ell]$, which is consistent iff $|g| \leq \ell$.

Combining the three cases,

$$\hat{\eta}_j = \begin{cases} (g - \ell)/h, & g > \ell, \\ 0, & |g| \leq \ell, \\ (g + \ell)/h, & g < -\ell, \end{cases}$$

which collapses to the single expression $\hat{\eta}_j = h^{-1} \text{sign}(g) \max\{0, |g| - \ell\}$. Restoring $h = \hat{H}_{jj}$, $g = \hat{g}_j$, $\ell = \lambda d_j^2/2$ yields the first form of (4). The second form is obtained by factoring out $\hat{H}_{jj}$ using $\hat{\eta}_j^{\text{OLS}} = \hat{g}_j/\hat{H}_{jj}$ and noting that sign is invariant under multiplication by the positive scalar $\hat{H}_{jj}$.

Finally, restating the minimizer in terms of the inverse-variance unpenalized weight $\hat{\eta}_j^{\text{OLS}}$ and the data-adaptive threshold $\lambda d_j^2/(2\hat{H}_{jj})$ emphasizes the soft-thresholding interpretation: sources whose unpenalized weight is dominated by the squared target–source gap are exactly screened out. This is the high-dimensional counterpart of the selection behavior observed in the coordinate-wise structure of the FACE aggregation rule of Han et al. [2025], which, however, retains a non-trivial cross-source coupling through $\hat{h}_{kt}(\boldsymbol{\eta})$ (the analogue of H^{OR}) and therefore does not admit a closed form in general. $\square$

Remark B.1 (When does condition (3) hold?). *Condition (3) is not generic: because $\xi_{t,s_j}^{\text{OR}}(D)$ is supported on the shared target sample, its covariance with $\xi_{t,s_k}^{\text{OR}}(D)$ depends on the correlation structure of the outcome models fitted with different pairwise nuisance targets $\boldsymbol{\alpha}_{t,s_j}^{1,*}$ and $\boldsymbol{\alpha}_{t,s_k}^{1,*}$. The condition holds in two natural limiting regimes: (i) when the per-source residual variances dominate the shared-target cross-covariances so that the off-diagonal entries of H^{OR} are negligible compared to the diagonal entries of H^{IPW} ; (ii) when the pairwise OR targets coincide asymptotically, in which case the shared-target covariance collapses into a rank-one structure that can be absorbed into the intercept. Outside these regimes, Corollary B.2 provides a first-order approximation whose accuracy depends on the magnitude of the off-diagonal entries of H^{OR} ; the general solution, both in FACE and in FACE-HD, requires solving the full K -dimensional penalized QP as in Equation (22).*

B.3 Penalized Variance Estimation

Define the empirical loss function as the estimated asymptotic variance of the aggregated estimator:

$$\hat{L}(\boldsymbol{\eta}) = N_{\text{all}} \cdot \widehat{\text{Var}}(\hat{\mu}_{\text{agg}}^1(\boldsymbol{\eta}))$$

which is a quadratic function of $\boldsymbol{\eta}$ (equivalently, N_{all} times the variance expression in Equation (22) in the main manuscript):

$$\hat{L}(\boldsymbol{\eta}) = \boldsymbol{\eta}^T \hat{H} \boldsymbol{\eta} + \hat{g}^T \boldsymbol{\eta} + \hat{c}$$

where $\hat{H}$, $\hat{g}$, and $\hat{c}$ are the estimated Hessian, gradient, and constant terms derived from the variance components in Section 4.2 of the main manuscript.

Lemma B.3 (Consistency of Variance/Covariance Components). *Assume Assumptions A.1 and A.2, and that K is fixed. Let H, g, c denote the population analogues of $\hat{H}, \hat{g}, \hat{c}$ defined by replacing empirical averages with expectations and using the population influence functions (with the same cross-site covariance terms induced by the shared target sample). Then:*

$$\|\hat{H} - H\|_{\text{op}} = O_p(N_{\text{all}}^{-1/2}), \quad \|\hat{g} - g\|_2 = O_p(N_{\text{all}}^{-1/2}), \quad |\hat{c} - c| = O_p(N_{\text{all}}^{-1/2}),$$

and consequently, for any fixed compact set $\mathcal{B} \subset \mathbb{R}^K$,

$$\sup_{\boldsymbol{\eta} \in \mathcal{B}} |\hat{L}(\boldsymbol{\eta}) - L^*(\boldsymbol{\eta})| = O_p(N_{\text{all}}^{-1/2}).$$

Proof. We prove entrywise $\sqrt{N_{\text{all}}}$ -consistency and then convert to the stated matrix/vector norms using that K is fixed.

Step 1 (Generic moment form). Each element of $\widehat{H}$, $\widehat{g}$ and $\widehat{c}$ is a (finite) linear combination of empirical averages of the form

$$\widehat{M}_r = \frac{1}{N_r} \sum_{i \in \mathcal{D}_r} f(D_i; \widehat{\vartheta}^{(-\text{fold}(i))}), \quad r \in \{t, s_1, \dots, s_K\},$$

where f is a product of two empirical influence-function components (including the cross-site covariance terms computed on the *same* target sample), and $\widehat{\vartheta}^{(-\text{fold}(i))}$ denotes the cross-fitted nuisance estimate used to evaluate the fold containing i . Let M_r be the population analogue obtained by replacing the sample average with $\mathbb{E}_r[\cdot]$ and $\widehat{\vartheta}^{(-\text{fold}(i))}$ with the population target ϑ^* .

Step 2 (Conditional LLN and plug-in stability). By cross-fitting, conditional on the nuisance-training samples, the summands $\{f(D_i; \widehat{\vartheta}^{(-\text{fold}(i))}) : i \in \mathcal{D}_r\}$ are independent within each site r , with conditional mean $\mathbb{E}_r[f(D; \widehat{\vartheta}^{(-\text{fold}(D))}) \mid \{\mathcal{D}^{(-k)}\}]$. Assumption A.2(b) gives a finite $(1 + \delta)$ moment for $f(D; \vartheta)$ uniformly in a neighborhood of ϑ^* (since it is a product of two $(2 + \delta)$ -moment IF components), hence the conditional variance is finite and a conditional LLN/CLT implies

$$\widehat{M}_r - \mathbb{E}_r[f(D; \widehat{\vartheta}^{(-\text{fold}(D))}) \mid \{\mathcal{D}^{(-k)}\}] = O_p(N_r^{-1/2}).$$

Next, by Lemma A.8, $\widehat{\vartheta}^{(-k)} \rightarrow_p \vartheta^*$, and by Lemma A.10 (Part 2) the mapping $\vartheta \mapsto f(D; \vartheta)$ is locally L_2 -Lipschitz. Therefore,

$$\mathbb{E}_r[f(D; \widehat{\vartheta}^{(-\text{fold}(D))})] = \mathbb{E}_r[f(D; \vartheta^*)] + o(1),$$

and combining the two displays yields $\widehat{M}_r - M_r = O_p(N_r^{-1/2})$. Under the site-ratio condition $N_r/N_{\text{all}} \rightarrow \rho_r > 0$, this is $O_p(N_{\text{all}}^{-1/2})$.

Step 3 (From entrywise to matrix/vector norms). Since $\widehat{H}$ has fixed dimension $K \times K$ and $\widehat{g}$ has fixed dimension K , the preceding bound applies to each entry, hence

$$\|\widehat{H} - H\|_{\text{op}} \leq C_K \max_{j,k} |\widehat{H}_{jk} - H_{jk}| = O_p(N_{\text{all}}^{-1/2}), \quad \|\widehat{g} - g\|_2 \leq \sqrt{K} \max_j |\widehat{g}_j - g_j| = O_p(N_{\text{all}}^{-1/2}),$$

Finally, $\widehat{c}$ is a finite linear combination of the same sample averages $\{\widehat{M}_r\}$ (coming from the constant terms in the variance expansion), so the same entrywise $O_p(N_{\text{all}}^{-1/2})$ bound implies $|\widehat{c} - c| = O_p(N_{\text{all}}^{-1/2})$.

Step 4 (Uniform convergence of the quadratic form). For η in a fixed compact set $\mathcal{B} \subset \mathbb{R}^K$,

$$|\widehat{L}(\eta) - L^*(\eta)| \leq \|\widehat{H} - H\|_{\text{op}} \|\eta\|_2^2 + \|\widehat{g} - g\|_2 \|\eta\|_2 + |\widehat{c} - c|,$$

and the right-hand side is $O_p(N_{\text{all}}^{-1/2})$ uniformly over $\eta \in \mathcal{B}$ because $\sup_{\eta \in \mathcal{B}} \|\eta\|_2 < \infty$. $\square$

The penalized objective from Equation (22) is:

$$\widehat{\eta} = \arg \min_{\eta} \left\{ \widehat{L}(\eta) + \lambda \sum_{j=1}^K |\eta_j| \left(\widehat{\mu}_{ot}^1 - \widehat{\mu}_{t,s_j}^1 \right)^2 \right\} \quad (6)$$

where $\lambda > 0$ is a tuning parameter that depends on N_{all} .

Remark: The penalty term $(\widehat{\mu}_{ot}^1 - \widehat{\mu}_{t,s_j}^1)^2$ measures the discrepancy between the target-only and source-assisted estimators. Large discrepancies indicate potential covariate shift or model misspecification, and the penalty drives the corresponding η_j toward zero.

B.4 Informative Source Sites

Let $\mu^1 = \mathbb{E}_t[Y(1)]$ be the true target-site potential outcome mean.

Define the set of *informative* source sites as:

$$\mathcal{S}^* = \left\{ j \in \{1, \dots, K\} : \text{plim } \widehat{\mu}_{t,s_j}^1 = \mu^1 \right\}$$

i.e., sites where the source-assisted estimator is consistent for the target parameter μ^1 .

Define the oracle-restricted space:

$$\mathbb{R}_{\mathcal{S}^*} := \{\eta \in \mathbb{R}^K : \eta_j = 0 \text{ for all } j \notin \mathcal{S}^*\}.$$

Let η^* be the oracle optimal weights that minimize the population asymptotic variance over $\mathbb{R}_{\mathcal{S}^*}$:

$$\eta^* = \arg \min_{\eta \in \mathbb{R}_{\mathcal{S}^*}} L^*(\eta), \quad L^*(\eta) := \text{Var} \left(\sqrt{N_{\text{all}}} (\hat{\mu}_{\text{agg}}^1(\eta) - \mu^1) \right).$$

Assumption B.1 (Aggregation Regularity). (a) (Sufficient Informative Sources + Local Strong Convexity) The set $\mathcal{S}^*$ is non-empty, i.e., there exists at least one informative source site. Moreover, the population aggregation objective is locally strictly convex on the oracle subspace: the Hessian of $L^*(\eta)$ restricted to $\mathbb{R}_{\mathcal{S}^*}$, denoted $H_{\mathcal{S}^*}$, is positive definite (equivalently, η^* is locally unique). (b) (Separation / “ β -min” for biased sites) Define

$$b_{0,N} := \min_{j \notin \mathcal{S}^*} \left| \text{plim}(\hat{\mu}_{t,s_j}^1) - \mu^1 \right|.$$

Assume $\lambda b_{0,N}^2 \rightarrow \infty$. A special case is fixed separation: $b_{0,N} \geq b_0 > 0$. (c) (Penalty Growth) The penalty parameter satisfies $\lambda \rightarrow \infty$ and $\lambda/\sqrt{N_{\text{all}}} \rightarrow 0$. (d) (Fixed K) The number of source sites K is fixed.

Remark B.2 (When separation fails / local alternatives). Assumption B.1(b) is used only for support recovery (forcing $\hat{\eta}_j = 0$ for biased sites). If some biased sites are only weakly biased in the sense that $b_{0,N} \rightarrow 0$, then selection consistency may fail. In that regime, the aggregated estimator remains asymptotically linear for any deterministic η (Lemma B.4), and the additional first-order bias is controlled by

$$\left| \sum_{j \notin \mathcal{S}^*} \eta_j \{ \text{plim}(\hat{\mu}_{t,s_j}^1) - \mu^1 \} \right|,$$

so if any included weakly biased sites satisfy $b_{j,N} = o(N_{\text{all}}^{-1/2})$, their contribution is still negligible at $\sqrt{N_{\text{all}}}$ scale.

B.5 Technical Lemmas for Aggregation

Lemma B.4 (Influence Function Representation). Under Assumptions A.1-A.3 and B.1(a), the aggregated estimator admits the asymptotic linear representation:

$$\sqrt{N_{\text{all}}} (\hat{\mu}_{\text{agg}}^1(\eta^*) - \mu^1) = \frac{1}{\sqrt{N_{\text{all}}}} \sum_{i=1}^{N_{\text{all}}} \varphi_{\text{agg},i}(\eta^*) + o_p(1)$$

where $\varphi_{\text{agg},i}(\eta^*)$ is the aggregated influence function:

$$\varphi_{\text{agg},i}(\eta^*) = \left(1 - \sum_{j \in \mathcal{S}^*} \eta_j^* \right) \varphi_{ot,i} + \sum_{j \in \mathcal{S}^*} \eta_j^* \varphi_{t,s_j,i},$$

with $\varphi_{t,s_j,i}$ denoting the (rescaled) IF contribution of source-assisted estimator j under $\sqrt{N_{\text{all}}}$ scaling.

Convention: Each $\varphi_{ot,i}$ (resp. $\varphi_{t,s_j,i}$) is understood as zero for observations not in the target data $\mathcal{D}_t$ (resp. not in $\mathcal{D}_t \cup \mathcal{D}_{s_j}$), and includes the constant rescaling needed to convert its natural $\sqrt{N_t}$ (resp. $\sqrt{N_{t,s_j}}$) linear representation into the common $\sqrt{N_{\text{all}}}$ scaling.

Proof. By Theorem A.3, for each fixed $j \in \mathcal{S}^*$, there is an asymptotic linear representation under pairwise scaling $N_{t,s_j} := N_t + N_{s_j}$:

$$\sqrt{N_{t,s_j}} (\hat{\mu}_{t,s_j}^1 - \text{plim}(\hat{\mu}_{t,s_j}^1)) = \frac{1}{\sqrt{N_{t,s_j}}} \sum_{i=1}^{N_{\text{all}}} \bar{\varphi}_{t,s_j,i} + o_p(1),$$

where $\bar{\varphi}_{t,s_j,i}$ is understood as zero for observations not in $\mathcal{D}_t \cup \mathcal{D}_{s_j}$. Since $N_{t,s_j}/N_{\text{all}} \rightarrow c_j \in (0, 1)$, the above is equivalent to

$$\sqrt{N_{\text{all}}} (\hat{\mu}_{t,s_j}^1 - \text{plim}(\hat{\mu}_{t,s_j}^1)) = \frac{1}{\sqrt{N_{\text{all}}}} \sum_{i=1}^{N_{\text{all}}} \varphi_{t,s_j,i} + o_p(1)$$

for a rescaled IF $\varphi_{t,s_j,i}$. The target-only estimator has the analogous form:

$$\sqrt{N_{\text{all}}}(\hat{\mu}_{ot}^1 - \mu^1) = \frac{1}{\sqrt{N_{\text{all}}}} \sum_{i=1}^{N_{\text{all}}} \varphi_{ot,i} + o_p(1).$$

For $j \in \mathcal{S}^*$, $\text{plim}(\hat{\mu}_{t,s_j}^1) = \mu^1$ by definition.

By linearity, the aggregated estimator with fixed $\eta = \eta^*$ satisfies:

$$\sqrt{N_{\text{all}}}(\hat{\mu}_{\text{agg}}^1(\eta^*) - \mu^1) = \frac{1}{\sqrt{N_{\text{all}}}} \sum_{i=1}^{N_{\text{all}}} \varphi_{\text{agg},i}(\eta^*) + o_p(1).$$

The remainder remains $o_p(1)$ because K is fixed. $\square$

Lemma B.5 (Selection Consistency). *Under Assumptions A.1-A.3 and B.1, the penalized estimator $\hat{\eta}$ from (6) satisfies:*

1. **(Rate)** $\|\hat{\eta} - \eta^*\|_2 = O_p(N_{\text{all}}^{-1/2})$
2. **(Selection)** $\hat{\eta}_j = 0$ for all $j \notin \mathcal{S}^*$ with probability approaching 1.

Proof. We give a self-contained argument; it parallels the FACE proof in Han et al. [2025] (Lemma S5) but the objective here is an explicit quadratic form. Recall $\hat{L}(\eta) = \eta^T \hat{H} \eta + \hat{g}^T \eta + \hat{c}$ and its population analogue $L^*(\eta) = \eta^T H \eta + g^T \eta + c$.

Part 1 (Rate on the oracle subspace). Define the oracle-restricted estimator

$$\tilde{\eta} = \arg \min_{\eta \in \mathbb{R}_{\mathcal{S}^*}} \left\{ \hat{L}(\eta) + \lambda \sum_{j=1}^K |\eta_j| (\hat{\mu}_{ot}^1 - \hat{\mu}_{t,s_j}^1)^2 \right\}.$$

On $\mathbb{R}_{\mathcal{S}^*}$, only coordinates $j \in \mathcal{S}^*$ can be nonzero, so the penalty term reduces to $\sum_{j \in \mathcal{S}^*} |\eta_j| (\hat{\mu}_{ot}^1 - \hat{\mu}_{t,s_j}^1)^2$. By Theorem A.3 and the site-ratio regularity, for each fixed $j \in \mathcal{S}^*$, $\hat{\mu}_{ot}^1 - \hat{\mu}_{t,s_j}^1 = O_p(N_{\text{all}}^{-1/2})$, hence $(\hat{\mu}_{ot}^1 - \hat{\mu}_{t,s_j}^1)^2 = O_p(N_{\text{all}}^{-1})$ and therefore the penalty contribution on $\mathbb{R}_{\mathcal{S}^*}$ is $O_p(\lambda/N_{\text{all}}) = o_p(N_{\text{all}}^{-1/2})$ under Assumption B.1(c).

Next, by Lemma B.3, the quadratic coefficients satisfy $\|\hat{H} - H\|_{\text{op}} = O_p(N_{\text{all}}^{-1/2})$ and $\|\hat{g} - g\|_2 = O_p(N_{\text{all}}^{-1/2})$. Let $H_{\mathcal{S}^*}$ and $\hat{H}_{\mathcal{S}^*}$ denote the principal submatrices corresponding to $\mathcal{S}^*$, and similarly for g . Because K is fixed, $H_{\mathcal{S}^*}$ is a fixed-dimensional positive definite matrix (local strong convexity of L^* around η^*), hence $\|H_{\mathcal{S}^*}^{-1}\|_{\text{op}} < \infty$ and $\|\hat{H}_{\mathcal{S}^*}^{-1}\|_{\text{op}} = O_p(1)$. More explicitly, by Weyl's inequality and Lemma B.3, $\lambda_{\min}(\hat{H}_{\mathcal{S}^*}) \geq \lambda_{\min}(H_{\mathcal{S}^*}) - \|\hat{H}_{\mathcal{S}^*} - H_{\mathcal{S}^*}\|_{\text{op}}$, which is bounded away from 0 w.p.a.1, so $\hat{H}_{\mathcal{S}^*}$ is invertible and $\|\hat{H}_{\mathcal{S}^*}^{-1}\|_{\text{op}} = O_p(1)$. Ignoring the $o_p(N_{\text{all}}^{-1/2})$ penalty perturbation, the minimizers of $\hat{L}$ and L^* on $\mathbb{R}_{\mathcal{S}^*}$ satisfy the first-order conditions $2\hat{H}_{\mathcal{S}^*}\eta + \hat{g}_{\mathcal{S}^*} = 0$ and $2H_{\mathcal{S}^*}\eta + g_{\mathcal{S}^*} = 0$, so

$$\eta_{\mathcal{S}^*}^* = -\frac{1}{2}H_{\mathcal{S}^*}^{-1}g_{\mathcal{S}^*}, \quad \tilde{\eta}_{\mathcal{S}^*} = -\frac{1}{2}\hat{H}_{\mathcal{S}^*}^{-1}\hat{g}_{\mathcal{S}^*} + o_p(N_{\text{all}}^{-1/2}).$$

To make the perturbation step explicit, write

$$\tilde{\eta}_{\mathcal{S}^*} - \eta_{\mathcal{S}^*}^* = -\frac{1}{2} \left(\hat{H}_{\mathcal{S}^*}^{-1}\hat{g}_{\mathcal{S}^*} - H_{\mathcal{S}^*}^{-1}g_{\mathcal{S}^*} \right) + o_p(N_{\text{all}}^{-1/2}).$$

Then

$$\hat{H}_{\mathcal{S}^*}^{-1}\hat{g}_{\mathcal{S}^*} - H_{\mathcal{S}^*}^{-1}g_{\mathcal{S}^*} = \hat{H}_{\mathcal{S}^*}^{-1}(\hat{g}_{\mathcal{S}^*} - g_{\mathcal{S}^*}) + (\hat{H}_{\mathcal{S}^*}^{-1} - H_{\mathcal{S}^*}^{-1})g_{\mathcal{S}^*}.$$

The first term is $O_p(N_{\text{all}}^{-1/2})$ since $\|\hat{H}_{\mathcal{S}^*}^{-1}\|_{\text{op}} = O_p(1)$ and $\|\hat{g} - g\|_2 = O_p(N_{\text{all}}^{-1/2})$. For the second term, use $\hat{H}_{\mathcal{S}^*}^{-1} - H_{\mathcal{S}^*}^{-1} = \hat{H}_{\mathcal{S}^*}^{-1}(H_{\mathcal{S}^*} - \hat{H}_{\mathcal{S}^*})H_{\mathcal{S}^*}^{-1}$, so

$$\|\hat{H}_{\mathcal{S}^*}^{-1} - H_{\mathcal{S}^*}^{-1}\|_{\text{op}} \leq \|\hat{H}_{\mathcal{S}^*}^{-1}\|_{\text{op}} \|\hat{H} - H\|_{\text{op}} \|H_{\mathcal{S}^*}^{-1}\|_{\text{op}} = O_p(N_{\text{all}}^{-1/2}),$$

and hence the second term is also $O_p(N_{\text{all}}^{-1/2})$. Therefore $\|\tilde{\eta} - \eta^*\|_2 = O_p(N_{\text{all}}^{-1/2})$. Finally, since $\hat{\eta}$ agrees with $\tilde{\eta}$ w.p.a.1 by Part 2 below, the same rate holds for $\hat{\eta}$.

Part 2 (Selection of biased sites). By Part 1, $\tilde{\eta} = \eta^* + O_p(N_{\text{all}}^{-1/2})$, so there exists a fixed-radius neighborhood $\mathcal{B} := \{\eta : \|\eta - \eta^*\|_2 \leq r\}$ such that $P(\tilde{\eta} \in \mathcal{B}) \rightarrow 1$. For the full penalized minimizer, the

same neighborhood restriction applies w.p.a.1 once we show (below) that all biased coordinates are set to zero, i.e., $\hat{\eta} = \tilde{\eta}$ w.p.a.1. Thus, in the KKT bound we may work on $\mathcal{B}$.

For biased sites $j \notin \mathcal{S}^*$, let $b_{j,N} := |\text{plim}(\hat{\mu}_{t,s_j}^1) - \mu^1|$, so $b_{j,N} \geq b_{0,N}$. Because $\hat{\mu}_{t,s_j}^1 \xrightarrow{p} \text{plim}(\hat{\mu}_{t,s_j}^1)$ and $\hat{\mu}_{ot}^1 \xrightarrow{p} \mu^1$,

$$(\hat{\mu}_{ot}^1 - \hat{\mu}_{t,s_j}^1)^2 \xrightarrow{p} b_{j,N}^2 \geq b_{0,N}^2,$$

and thus $\lambda(\hat{\mu}_{ot}^1 - \hat{\mu}_{t,s_j}^1)^2 \rightarrow \infty$ under Assumption B.1(b).

The KKT condition for the penalized objective at coordinate j implies that if $\hat{\eta}_j = 0$ then

$$\left| \frac{\partial \hat{L}}{\partial \eta_j}(\hat{\eta}) \right| \leq \lambda(\hat{\mu}_{ot}^1 - \hat{\mu}_{t,s_j}^1)^2.$$

Since $\hat{L}(\eta) = \eta^T \hat{H} \eta + \hat{g}^T \eta + \hat{c}$ is quadratic,

$$\frac{\partial \hat{L}}{\partial \eta_j}(\eta) = 2(\hat{H} \eta)_j + \hat{g}_j,$$

so for any compact $\mathcal{B} \subset \mathbb{R}^K$,

$$\sup_{\eta \in \mathcal{B}} \left| \frac{\partial \hat{L}}{\partial \eta_j}(\eta) \right| \leq 2 \sup_{\eta \in \mathcal{B}} \|\hat{H} \eta\|_\infty + \|\hat{g}\|_\infty \leq 2 \|\hat{H}\|_{\text{op}} \sup_{\eta \in \mathcal{B}} \|\eta\|_2 + \|\hat{g}\|_2.$$

By Lemma B.3, $\|\hat{H}\|_{\text{op}} = O_p(1)$ and $\|\hat{g}\|_2 = O_p(1)$, hence $\sup_{\eta \in \mathcal{B}} |\partial \hat{L} / \partial \eta_j(\eta)| = O_p(1)$. Because the right-hand side diverges, the KKT inequality is satisfied for $\eta_j = 0$ w.p.a.1 and any minimizer must set $\hat{\eta}_j = 0$ for all $j \notin \mathcal{S}^*$. Therefore $\hat{\eta} \in \mathbb{R}_{\mathcal{S}^*}$ w.p.a.1, hence $\hat{\eta} = \tilde{\eta}$ w.p.a.1. $\square$

Lemma B.6 (Asymptotic Equivalence). *Under Assumptions A.1-A.3 and B.1:*

$$\sqrt{N_{\text{all}}} (\hat{\mu}_{\text{agg}}^1(\hat{\eta}) - \hat{\mu}_{\text{agg}}^1(\eta^*)) = o_p(1)$$

Proof. The difference is:

$$\sqrt{N_{\text{all}}} (\hat{\mu}_{\text{agg}}^1(\hat{\eta}) - \hat{\mu}_{\text{agg}}^1(\eta^*)) = \sum_{j=1}^K (\hat{\eta}_j - \eta_j^*) \cdot \sqrt{N_{\text{all}}} (\hat{\mu}_{t,s_j}^1 - \hat{\mu}_{ot}^1)$$

For $j \in \mathcal{S}^*$: By Lemma B.5(1), $|\hat{\eta}_j - \eta_j^*| = O_p(N_{\text{all}}^{-1/2})$. By Theorem A.3 with the same rescaling argument as in Lemma B.4, $\sqrt{N_{\text{all}}} (\hat{\mu}_{t,s_j}^1 - \hat{\mu}_{ot}^1) = O_p(1)$. Thus each term is $o_p(1)$.

For $j \notin \mathcal{S}^*$: By Lemma B.5(2), $\hat{\eta}_j = 0$ w.h.p., and $\eta_j^* = 0$ by definition (oracle excludes biased sites). Thus the difference is zero w.h.p. Because K is fixed, summing over j yields $o_p(1)$. $\square$

Remark B.3 (Growing number of sources K). *The arguments above use Assumption B.1(d) (fixed K) in two places: (i) to ensure the plug-in term $\sum_{j=1}^K (\hat{\eta}_j - \eta_j^*) \sqrt{N_{\text{all}}} (\hat{\mu}_{t,s_j}^1 - \hat{\mu}_{ot}^1)$ is $o_p(1)$, and (ii) to treat the variance/covariance estimation problem as finite-dimensional. If $K = K_N \rightarrow \infty$, then additional high-dimensional control is needed, e.g., a sufficient condition for Lemma B.6 is*

$$\|\hat{\eta} - \eta^*\|_1 \cdot \max_{1 \leq j \leq K} \left| \sqrt{N_{\text{all}}} (\hat{\mu}_{t,s_j}^1 - \hat{\mu}_{ot}^1) \right| = o_p(1),$$

which typically requires both a sparsity assumption on η^ and concentration bounds scaling with $\log K$.*

B.6 Main Results for Aggregation

Theorem B.7 (Asymptotic Normality of FACE-HD). *Suppose Assumptions A.1-A.3 and B.1 hold. For each informative source $s_j \in \mathcal{S}^*$, let $N_{t,s_j} = N_t + N_{s_j}$ and $s_{\text{tot},j} = s_{\alpha,j} + s_{\gamma,j}$ denote the pairwise total sparsity. If*

$$\max_{j \in \mathcal{S}^*} \frac{s_{\text{tot},j}^2 \log^2(p)}{N_{t,s_j}} \rightarrow 0,$$

(equivalently under fixed site ratios, the same condition with N_{all}), then:

$$\sqrt{N_{\text{all}}} (\hat{\mu}_{\text{agg}}^1 - \mu^1) \xrightarrow{d} \mathcal{N}(0, L^*(\eta^*))$$

where $L^*(\eta^*)$ is the minimum asymptotic variance achieved by the oracle aggregation.

Moreover, the variance estimator is consistent:

$$\hat{V} = \hat{L}(\hat{\eta}) \xrightarrow{p} L^*(\eta^*)$$

and the standardized statistic is asymptotically standard normal:

$$\sqrt{N_{\text{all}}/\hat{V}} (\hat{\mu}_{\text{agg}}^1 - \mu^1) \xrightarrow{d} \mathcal{N}(0, 1)$$

Proof. Combining Lemmas B.4, B.5, and B.6:

$$\sqrt{N_{\text{all}}} (\hat{\mu}_{\text{agg}}^1 - \mu^1) = \sqrt{N_{\text{all}}} (\hat{\mu}_{\text{agg}}^1(\eta^*) - \mu^1) + o_p(1) = \frac{1}{\sqrt{N_{\text{all}}}} \sum_{i=1}^{N_{\text{all}}} \varphi_{\text{agg},i}(\eta^*) + o_p(1)$$

As in Lemma A.5, the observations are independent with a multi-site structure and site proportions converging to constants. The aggregated influence function contributions $\{\varphi_{\text{agg},i}(\eta^*)\}_{i=1}^{N_{\text{all}}}$ are therefore independent with mean zero and finite variance $L^*(\eta^*)$. Under Assumption A.2(b), these contributions have a uniform $(2 + \delta)$ moment, so the same Lyapunov argument as in Lemma A.5 applies and yields:

$$\frac{1}{\sqrt{N_{\text{all}}}} \sum_{i=1}^{N_{\text{all}}} \varphi_{\text{agg},i}(\eta^*) \xrightarrow{d} \mathcal{N}(0, L^*(\eta^*))$$

For variance estimation consistency: By Lemma B.5(1), $\|\hat{\eta} - \eta^*\|_2 = O_p(N_{\text{all}}^{-1/2})$. By the uniform convergence of $\hat{L}$ to L^* and local Lipschitz continuity of the quadratic map $L^*(\eta) = \eta^T H \eta + g^T \eta + c$ in a neighborhood of η^* (e.g., on $\{\eta : \|\eta - \eta^*\|_2 \leq R\}$, $|L^*(\eta_1) - L^*(\eta_2)| \leq (2\|H\|_{\text{op}}R + \|g\|_2)\|\eta_1 - \eta_2\|_2$):

$$|\hat{L}(\hat{\eta}) - L^*(\eta^*)| \leq |\hat{L}(\hat{\eta}) - L^*(\hat{\eta})| + |L^*(\hat{\eta}) - L^*(\eta^*)| = O_p(N_{\text{all}}^{-1/2}) + O_p(N_{\text{all}}^{-1/2}) = o_p(1)$$

Finally, write

$$\sqrt{\frac{N_{\text{all}}}{\hat{V}}} (\hat{\mu}_{\text{agg}}^1 - \mu^1) = \left(\sqrt{\frac{L^*(\eta^*)}{\hat{V}}} \right) \cdot \left(\frac{1}{\sqrt{L^*(\eta^*)}} \sqrt{N_{\text{all}}} (\hat{\mu}_{\text{agg}}^1 - \mu^1) \right).$$

We already proved $\sqrt{N_{\text{all}}} (\hat{\mu}_{\text{agg}}^1 - \mu^1) \xrightarrow{d} \mathcal{N}(0, L^*(\eta^*))$, hence the second factor converges in distribution to $\mathcal{N}(0, 1)$. By variance consistency, $\hat{V} \xrightarrow{p} L^*(\eta^*)$, so the prefactor $\sqrt{L^*(\eta^*)/\hat{V}} \xrightarrow{p} 1$. Therefore the product converges in distribution to $\mathcal{N}(0, 1)$ by Slutsky's theorem. $\square$

Corollary B.8 (Asymptotic Normality of the Aggregated ATE). *Define the aggregated ATE estimator as*

$$\hat{\tau}_{\text{agg}} := \hat{\mu}_{\text{agg}}^1 - \hat{\mu}_{\text{agg}}^0,$$

where $\hat{\mu}_{\text{agg}}^0$ is constructed by applying the same aggregation procedure to the control-arm estimators $\{\hat{\mu}_{t,s_j}^0\}_{j=1}^K$ (with the corresponding nuisance estimates and variance/covariance components). Assume the analogues of Assumptions A.1–B.1 hold for both arms, so that both $\hat{\mu}_{\text{agg}}^1$ and $\hat{\mu}_{\text{agg}}^0$ admit asymptotic linear representations under $\sqrt{N_{\text{all}}}$ scaling. Then

$$\sqrt{N_{\text{all}}} (\hat{\tau}_{\text{agg}} - \tau) \xrightarrow{d} \mathcal{N}(0, V_{\tau}^*), \quad V_{\tau}^* := \text{Var}(\varphi_{\text{agg}}^1(D) - \varphi_{\text{agg}}^0(D)),$$

where φ_{agg}^1 and φ_{agg}^0 denote the aggregated influence functions (oracle-level) for the treated and control aggregated estimators, respectively.

Proposition B.9 (Efficiency Gain). *Under Assumptions A.1–A.3 and B.1, the FACE-HD estimator is asymptotically at least as efficient as the target-only estimator:*

$$L^*(\eta^*) \leq L^*(0)$$

with strict inequality whenever there exists $j \in \mathcal{S}^*$ such that

$$\text{Var}(\varphi_{t,s_j} - \varphi_{ot}) > 0 \quad \text{and} \quad \text{Cov}(\varphi_{ot}, \varphi_{t,s_j} - \varphi_{ot}) \neq 0.$$

Proof. Since η^* minimizes $L^*(\eta)$ and $\eta = 0$ is feasible, we have $L^*(\eta^*) \leq L^*(0)$.

For strict improvement, consider one informative source $j \in \mathcal{S}^*$. The optimal projection is:

$$\eta_j^* = -\frac{\text{Cov}(\varphi_{ot}, \varphi_{t,s_j} - \varphi_{ot})}{\text{Var}(\varphi_{t,s_j} - \varphi_{ot})}$$

If the covariance is non-zero and the denominator is positive, the minimizing weight is non-zero and strictly decreases variance relative to $\eta = 0$, so $L^*(\eta^*) < L^*(0)$. $\square$

References

- Peter Bühlmann and Sara Van De Geer. *Statistics for high-dimensional data: methods, theory and applications*. Springer Science & Business Media, 2011.
- Larry Han, Jue Hou, Kelly Cho, Rui Duan, and Tianxi Cai. Federated adaptive causal estimation (face) of target treatment effects. *Journal of the American Statistical Association*, 120(551):1503–1516, 2025.
- Jue Hou, Rajarshi Mukherjee, and Tianxi Cai. Efficient and robust semi-supervised estimation of average treatment effect with partially annotated treatment and response. *Journal of Machine Learning Research*, 26(40):1–77, 2025.
- Sahand N Negahban, Pradeep Ravikumar, Martin J Wainwright, and Bin Yu. A unified framework for high-dimensional analysis of m-estimators with decomposable regularizers. *Statistical Science*, 27(4): 538–557, 2012.
- Zhiqiang Tan. Model-assisted inference for treatment effects using regularized calibrated estimation with high-dimensional data. *The Annals of Statistics*, 48(2):811–837, 2020.